

Movie_Audience_Score_Prediction

May 29, 2026

1 Milestone 1: Data Selection and EDA.

It is time to start using what you have learned throughout the first half of this course by developing an original data mining project. This week you will develop the project idea and do some data exploration/graphical analysis. You will continue working on and updating this project for the remainder of the term.

The first step is coming up with an idea – arguably one of the hardest steps! Identify an original business problem for your project that can be solved with an appropriate model. By a business problem, it is meant that you should work on a problem where there is a good reason to solve it. There should be some organization or company that would find the solution to the problem useful. There are lots of ideas available online through Kaggle and other sources, but your idea should have a unique spin on it. The second step is locating your data. This can come from a variety of sources, e.g., Kaggle, your job, a website, API, etc. Feel free to reach out to your instructor if you are not sure if your idea and data are suitable. You may need to adjust your idea on the availability of data.

Begin Milestone 1 with a 250-500-word narrative describing your original idea for the analysis/model building business problem. Clearly identify the problem you will address and the target for your model. Then, do a graphical analysis creating a minimum of four graphs. Label your graphs appropriately and explain/analyze the information provided by each graph. Your analysis should begin to answer the question(s) you are addressing. Write a short overview/conclusion of the insights gained from your graphical analysis.

```
[1]: # Load the necessary libraries.  
import pandas as pd  
import matplotlib.pyplot as plt
```

The datasets used in this notebook are from this website:

<https://www.kaggle.com/datasets/stefanoleone992/rotten-tomatoes-movies-and-critic-reviews-dataset>

```
[2]: # Load the provided data files.  
critic_reviews_df = pd.read_csv("rotten_tomatoes_critic_reviews.csv")  
movies_df = pd.read_excel("rotten_tomatoes_movies.xlsx")  
  
[3]: # Display the first few rows of critics review to understand structure.  
critic_reviews_df.head()
```

```
[3]: rotten_tomatoes_link    critic_name top_critic    publisher_name \
0      m/0814255  Andrew L. Urban      False      Urban Cinefile
1      m/0814255    Louise Keller      False      Urban Cinefile
2      m/0814255                NaN      False      FILMINK (Australia)
3      m/0814255    Ben McEachen      False      Sunday Mail (Australia)
4      m/0814255    Ethan Alter        True      Hollywood Reporter

      review_type review_score review_date \
0      Fresh      NaN    2010-02-06
1      Fresh      NaN    2010-02-06
2      Fresh      NaN    2010-02-09
3      Fresh      3.5/5    2010-02-09
4      Rotten      NaN    2010-02-10

      review_content
0  A fantasy adventure that fuses Greek mythology...
1  Uma Thurman as Medusa, the gorgon with a coiff...
2  With a top-notch cast and dazzling special eff...
3  Whether audiences will get behind The Lightnin...
4  What's really lacking in The Lightning Thief i...
```

```
[4]: # Dimensions of the critic_reviews DataFrame.
critic_reviews_df.shape
```

```
[4]: (1130017, 8)
```

```
[5]: # Display the first few rows movies_df to understand structure.
movies_df.head()
```

```
[5]: rotten_tomatoes_link \
0      m/0814255
1      m/0878835
2      m/10
3      m/1000013-12_angry_men
4  m/1000079-20000_leagues_under_the_sea

      movie_title \
0  Percy Jackson & the Olympians: The Lightning T...
1      Please Give
2      10
3      12 Angry Men (Twelve Angry Men)
4      20,000 Leagues Under The Sea

      movie_info \
0  Always trouble-prone, the life of teenager Per...
1  Kate (Catherine Keener) and her husband Alex (...
2  A successful, middle-aged Hollywood songwriter...
```

3 Following the closing arguments in a murder tr...
 4 In 1866, Professor Pierre M. Aronnax (Paul Luk...

	critics_consensus	content_rating	\
0	Though it may seem like just another Harry Pot...	PG	
1	Nicole Holofcener's newest might seem slight i...	R	
2	Blake Edwards' bawdy comedy may not score a pe...	R	
3	Sidney Lumet's feature debut is a superbly wri...	NR	
4	One of Disney's finest live-action adventures,...	G	

	genres	directors	\
0	Action & Adventure, Comedy, Drama, Science Fic...	Chris Columbus	
1	Comedy	Nicole Holofcener	
2	Comedy, Romance	Blake Edwards	
3	Classics, Drama	Sidney Lumet	
4	Action & Adventure, Drama, Kids & Family	Richard Fleischer	

	authors	\
0	Craig Titley, Chris Columbus, Rick Riordan	
1	Nicole Holofcener	
2	Blake Edwards	
3	Reginald Rose	
4	Earl Felton	

	actors	original_release_date	\
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12	
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30	
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05	
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13	
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01	

	production_company	tomatometer_status	tomatometer_rating	\
0	20th Century Fox	Rotten	49.0	
1	Sony Pictures Classics	Certified-Fresh	87.0	
2	Waner Bros.	Fresh	67.0	
3	Criterion Collection	Certified-Fresh	100.0	
4	Disney	Fresh	89.0	

	tomatometer_count	audience_status	audience_rating	audience_count	\
0	149.0	Spilled	53.0	254421.0	
1	142.0	Upright	64.0	11574.0	
2	24.0	Spilled	53.0	14684.0	
3	54.0	Upright	97.0	105386.0	
4	27.0	Upright	74.0	68918.0	

	tomatometer_top_critics_count	tomatometer_fresh_critics_count	\
0	43	73	

1	44	123
2	2	16
3	6	54
4	5	24

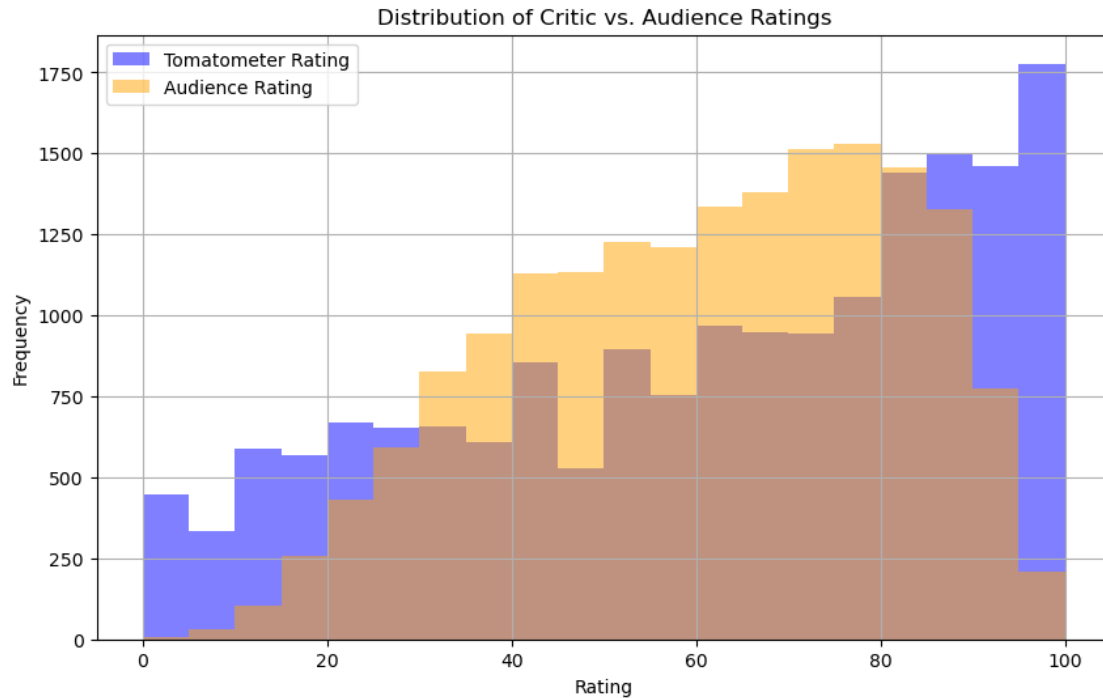
	tomatometer_rotten_critics_count
0	76
1	19
2	8
3	0
4	3

[5 rows x 22 columns]

```
[6]: # Dimensions of the movies_df DataFrame.
movies_df.shape
```

```
[6]: (17712, 22)
```

```
[7]: # 1. Distribution of Critic vs. Audience Ratings
plt.figure(figsize=(10, 6))
plt.hist(movies_df['tomatometer_rating'], bins=20, alpha=0.5,
        label='Tomatometer Rating', color='blue')
plt.hist(movies_df['audience_rating'], bins=20, alpha=0.5, label='Audience
        Rating', color='orange')
plt.title('Distribution of Critic vs. Audience Ratings')
plt.xlabel('Rating')
plt.ylabel('Frequency')
plt.legend()
plt.grid(True)
plt.show()
```



Understanding the Ratings: Critics vs. Audiences

This graph compares ratings from movie critics and general audiences across various films. On the bottom (the x-axis), we see the ratings ranging from 0 to 100. On the side (the y-axis), we see how often movies fall into different rating groups.

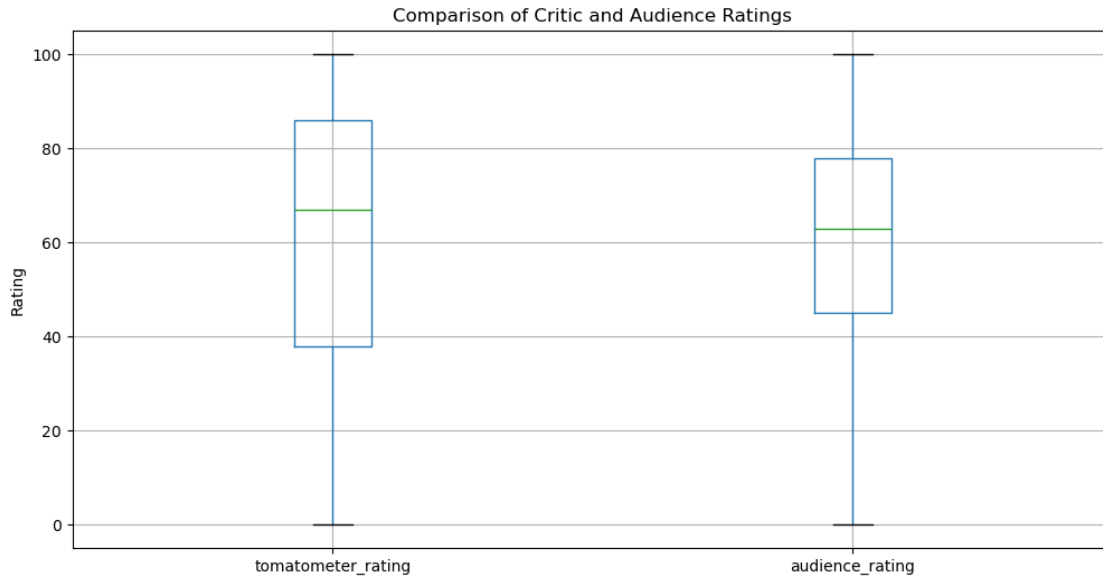
Analysis:

The way critics' ratings cluster in the middle suggests they take a more balanced approach to reviewing films. They focus on the technical aspects and the overall quality, which leads to fewer extreme scores.

In contrast, audience ratings show a wider range, indicating that general viewers are more likely to either enjoy or dislike a movie. Their ratings can be influenced by personal tastes and emotional connections, resulting in stronger opinions.

This graph highlights the differences between how critics and audiences view movies, reflecting their unique perspectives and preferences.

```
[8]: # 2. Box plot of Critic and Audience Rating.
plt.figure(figsize=(12, 6))
movies_df.boxplot(column=['tomatometer_rating', 'audience_rating'])
plt.title('Comparison of Critic and Audience Ratings')
plt.ylabel('Rating')
plt.grid(True)
plt.show()
```



Understanding the Box Plot of Critic and Audience Ratings

This box plot visually compares the ratings given by movie critics (Tomatometer) and regular audiences. The y-axis shows the ratings, which range from 0 to 100. The box plots provide important information about how these ratings are distributed.

Analysis:

Critics are generally more consistent in their ratings, while audience ratings vary more. This suggests that audience ratings might depend on a wider range of personal factors, like emotions or individual tastes.

The higher number of outliers in audience ratings indicates that people often have strong, divided opinions about certain movies—either loving or hating them. This could be due to niche or controversial content.

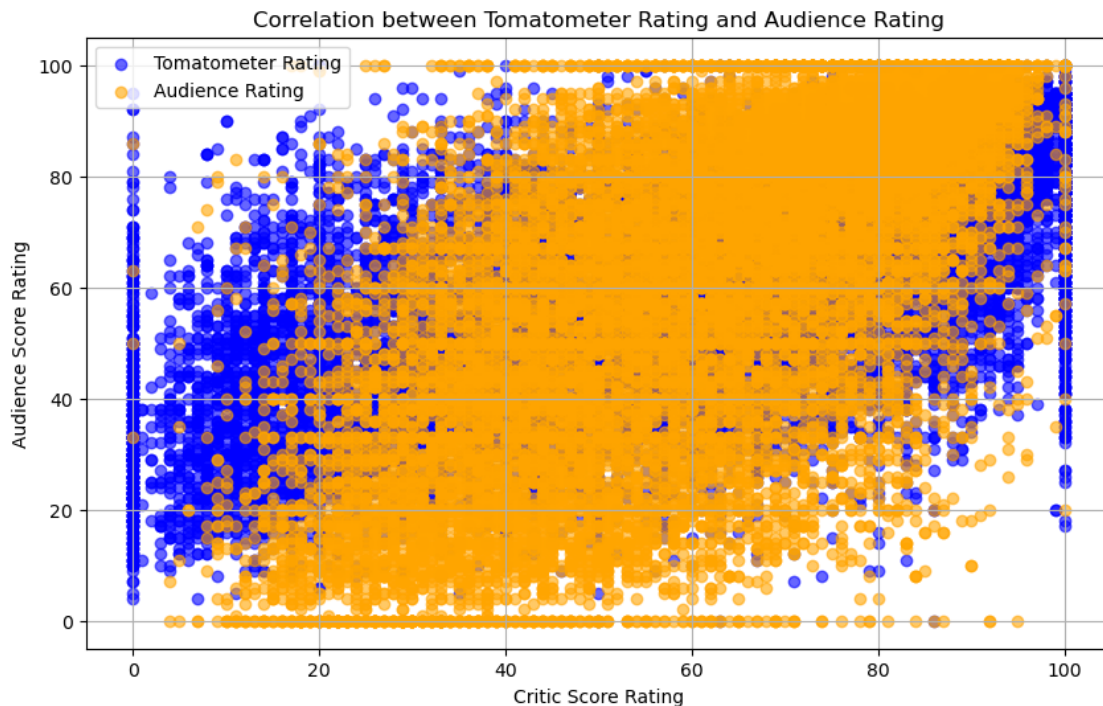
```
[9]: # 3. Correlation between Tomatometer Rating and Audience Rating
plt.figure(figsize=(10, 6))

# Scatter plot for Tomatometer Rating.
plt.scatter(movies_df['tomatometer_rating'], movies_df['audience_rating'],
            alpha=0.6, color='blue', label='Tomatometer Rating')

# Scatter plot for Audience Rating (adjusting for distinction).
plt.scatter(movies_df['audience_rating'], movies_df['tomatometer_rating'],
            alpha=0.6, color='orange', label='Audience Rating')

plt.title('Correlation between Tomatometer Rating and Audience Rating')
plt.xlabel('Critic Score Rating')
plt.ylabel('Audience Score Rating')
```

```
plt.grid(True)
plt.legend()
plt.show()
```



Understanding the Scatter Plot: Comparing Tomatometer Ratings and Audience Ratings

This scatter plot shows the relationship between ratings from movie critics (Tomatometer) and ratings from regular audiences for various films. Each point on the graph represents a different movie, with the critic score shown on the x-axis and the audience score on the y-axis. This helps us see whether critics and audiences tend to agree on movie ratings.

Analysis

The plot shows a general positive correlation between critic and audience ratings, but the scattered points indicate that this relationship isn't very strong. While critics and audiences often agree, there are notable differences in some cases.

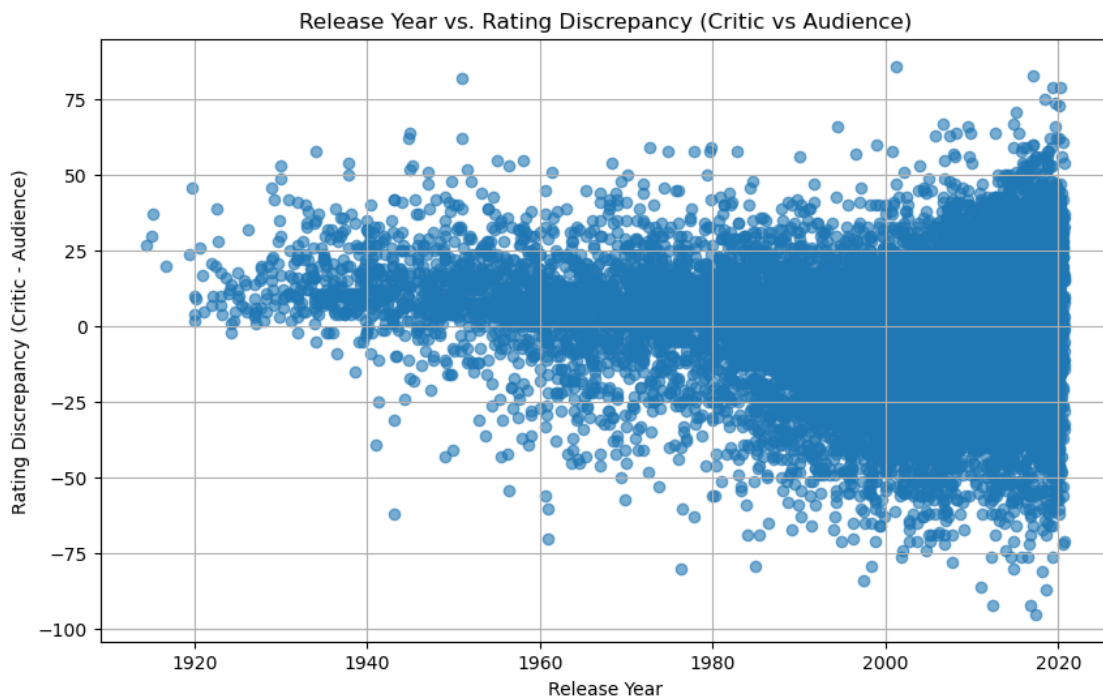
Movies where the critic and audience scores differ significantly are important to look at. These cases can provide insights into specific types of movies, genres, or qualities that lead to differing opinions.

This scatter plot begins to answer the question of how closely critic and audience ratings are related but also points to the need for further investigation to understand the differences in their ratings.

```
[10]: # 4. Release Year vs. Rating Discrepancy.

# Calculate rating discrepancy if not already done.
movies_df['rating_discrepancy'] = movies_df['tomatometer_rating'] -
    movies_df['audience_rating']

# Create a scatter plot for Release Year vs Rating Discrepancy
plt.figure(figsize=(10, 6))
plt.scatter(movies_df['original_release_date'],
    movies_df['rating_discrepancy'], alpha=0.6)
plt.title('Release Year vs. Rating Discrepancy (Critic vs Audience)')
plt.xlabel('Release Year')
plt.ylabel('Rating Discrepancy (Critic - Audience)')
plt.grid(True)
plt.show()
```



Understanding Release Year vs. Rating Discrepancy (Critic vs Audience)

This scatter plot shows how the year a movie was released relates to the difference between ratings from critics (Tomatometer) and ratings from the audience. Each dot on the graph represents a movie, with the x-axis showing the release year and the y-axis showing how much the ratings differ.

Analysis:

Rating discrepancy is the difference between the critic rating and the audience rating. If the value is positive, critics liked the movie more than the audience. If it's negative, it means the audience

enjoyed the movie more than the critics.

By looking at how the dots are spread over the years, we can see if there are any patterns in how critics and audiences rate movies differently.

If most points are close to zero, it suggests that critics and audiences generally agree on the ratings for those movies. A wider spread of points means there are more differences in opinions, indicating that audiences and critics might be more divided in their views.

2 Milestone 2: Data Preparation.

Now that you have created your idea, located data, and have started your graphical analysis, you will move on to the data preparation process of your project. After completing Milestone 2, your data should be ready for the model building/evaluation phase.

Here is a list of steps to consider performing in Milestone 2:

1. Drop any features that are not useful for your model building and explain why they are not useful.
2. Perform any data extraction/selection steps.
3. Transform features if necessary.
4. Engineer new useful features.
5. Deal with missing data (do not just drop rows or columns without justifying this).
6. Create dummy variables if necessary.

Explain your process at each step. You can use any methods/tools you think are most appropriate. Do what makes the most sense for your data/problem. This will vary greatly among different projects. Be careful to avoid data snooping in these steps.

It is important to note that these milestones are meant to keep you on track for the final project submission. At any point, you can pivot or modify your project as needed based on what you discover. These milestones are not final versions; they are drafts of the many steps you need to complete along the way.

```
[11]: # Load the necessary libraries.
import pandas as pd
import numpy as np
```

```
[12]: # Load both datasets.
critic_reviews_df = pd.read_csv("rotten_tomatoes_critic_reviews.csv")
movies_df = pd.read_excel("rotten_tomatoes_movies.xlsx")
```

3 Critic Reviews Dataset.

```
[13]: # Display the first few rows of critic_reviews_df to understand structure.
critic_reviews_df.head(60)
```

```
[13]: rotten_tomatoes_link      critic_name  top_critic \
0      m/0814255      Andrew L. Urban      False
1      m/0814255      Louise Keller      False
```

2	m/0814255	NaN	False
3	m/0814255	Ben McEachen	False
4	m/0814255	Ethan Alter	True
5	m/0814255	David Germain	True
6	m/0814255	Nick Schager	False
7	m/0814255	Bill Goodykoontz	True
8	m/0814255	Jordan Hoffman	False
9	m/0814255	Jim Schembri	True
10	m/0814255	Mark Adams	False
11	m/0814255	Roger Moore	True
12	m/0814255	David Jenkins	True
13	m/0814255	Joshua Tyler	False
14	m/0814255	Peter Paras	False
15	m/0814255	Vic Holtreman	False
16	m/0814255	MaryAnn Johanson	False
17	m/0814255	Mark Dujsik	False
18	m/0814255	Nell Minow	False
19	m/0814255	Mary Elizabeth Williams	True
20	m/0814255	Dustin Putman	False
21	m/0814255	Stephen Holden	True
22	m/0814255	Jeffrey M. Anderson	False
23	m/0814255	Elizabeth Weitzman	True
24	m/0814255	Kenneth Turan	True
25	m/0814255	Moirra MacDonald	True
26	m/0814255	James Rocchi	True
27	m/0814255	Stephen Whitty	True
28	m/0814255	David Plotz	True
29	m/0814255	Jeff Vice	False
30	m/0814255	Wesley Morris	True
31	m/0814255	Andrea Gronvall	True
32	m/0814255	Tasha Robinson	True
33	m/0814255	Rene Rodriguez	True
34	m/0814255	Kent Turner	False
35	m/0814255	Bill Gibron	False
36	m/0814255	Joe Williams	True
37	m/0814255	Cole Smithey	False
38	m/0814255	Michael Phillips	True
39	m/0814255	Matt Pais	False
40	m/0814255	Roger Ebert	True
41	m/0814255	Rich Cline	False
42	m/0814255	Fionnuala Halligan	False
43	m/0814255	Matthew Turner	False
44	m/0814255	Peter Debruge	True
45	m/0814255	Diva Velez	False
46	m/0814255	Christopher Smith	False
47	m/0814255	Brian Tallerico	False
48	m/0814255	Walter Chaw	False

49	m/0814255	Nick Starkey	False
50	m/0814255	Jenna Busch	False
51	m/0814255	Kurt Loder	False
52	m/0814255	David Frese	False
53	m/0814255	Amy Biancolli	True
54	m/0814255	Rick Groen	True
55	m/0814255	Adam Markovitz	True
56	m/0814255	Pete Hammond	False
57	m/0814255	Tom Long	True
58	m/0814255	Carrie R. Wheadon	False
59	m/0814255	Andrea Chase	False

	publisher_name	review_type	review_score	review_date	\
0	Urban Cinefile	Fresh	NaN	2010-02-06	
1	Urban Cinefile	Fresh	NaN	2010-02-06	
2	FILMINK (Australia)	Fresh	NaN	2010-02-09	
3	Sunday Mail (Australia)	Fresh	3.5/5	2010-02-09	
4	Hollywood Reporter	Rotten	NaN	2010-02-10	
5	Associated Press	Rotten	NaN	2010-02-10	
6	Slant Magazine	Rotten	1/4	2010-02-10	
7	Arizona Republic	Fresh	3.5/5	2010-02-10	
8	UGO	Fresh	B	2010-02-10	
9	The Age (Australia)	Fresh	3/5	2010-02-10	
10	Daily Mirror (UK)	Fresh	4/5	2010-02-10	
11	Orlando Sentinel	Rotten	2/4	2010-02-10	
12	Time Out	Rotten	2/5	2010-02-10	
13	CinemaBlend	Fresh	3/5	2010-02-11	
14	E! Online	Rotten	C	2010-02-11	
15	Screen Rant	Fresh	3.5/5	2010-02-11	
16	Flick Filosopher	Fresh	NaN	2010-02-11	
17	Mark Reviews Movies	Rotten	2.5/4	2010-02-11	
18	Beliefnet	Fresh	B	2010-02-11	
19	Salon.com	Rotten	NaN	2010-02-11	
20	TheFilmFile.com	Fresh	2.5/4	2010-02-11	
21	New York Times	Rotten	2/5	2010-02-11	
22	Combustible Celluloid	Rotten	1/4	2010-02-11	
23	New York Daily News	Fresh	3/5	2010-02-11	
24	Los Angeles Times	Rotten	2/5	2010-02-11	
25	Seattle Times	Rotten	2/4	2010-02-11	
26	MSN Movies	Fresh	3/5	2010-02-11	
27	Newark Star-Ledger	Fresh	3/4	2010-02-11	
28	Slate	Rotten	NaN	2010-02-11	
29	Deseret News (Salt Lake City)	Fresh	2/4	2010-02-11	
30	Boston Globe	Rotten	2/4	2010-02-11	
31	Chicago Reader	Rotten	NaN	2010-02-11	
32	AV Club	Rotten	C-	2010-02-11	
33	Miami Herald	Fresh	2.5/4	2010-02-11	

34	School Library Journal	Fresh	2.75/5	2010-02-11
35	Filmcritic.com	Rotten	3/5	2010-02-11
36	St. Louis Post-Dispatch	Fresh	2.5/4	2010-02-11
37	ColeSmithey.com	Fresh	B-	2010-02-11
38	Chicago Tribune	Rotten	2/4	2010-02-11
39	Metromix.com	Rotten	2/5	2010-02-11
40	Chicago Sun-Times	Fresh	3/4	2010-02-11
41	Shadows on the Wall	Fresh	3/5	2010-02-11
42	Screen International	Fresh	NaN	2010-02-11
43	ViewLondon	Fresh	3/5	2010-02-11
44	Variety	Fresh	NaN	2010-02-11
45	TheDivaReview.com	Rotten	2.5/5	2010-02-12
46	Bangor Daily News (Maine)	Rotten	C	2010-02-12
47	Movie Retriever	Rotten	NaN	2010-02-12
48	Film Freak Central	Rotten	1/4	2010-02-12
49	Premiere Magazine	Rotten	1/4	2010-02-12
50	Newsarama	Rotten	5.5/10	2010-02-12
51	MTV	Rotten	1/5	2010-02-12
52	Kansas City Star	Rotten	2/4	2010-02-12
53	Houston Chronicle	Fresh	3/4	2010-02-12
54	Globe and Mail	Rotten	2/4	2010-02-12
55	Entertainment Weekly	Rotten	C+	2010-02-12
56	Boxoffice Magazine	Rotten	2.5/5	2010-02-12
57	Detroit News	Rotten	D+	2010-02-12
58	Common Sense Media	Rotten	2/5	2010-02-12
59	Killer Movie Reviews	Fresh	3/5	2010-02-12

review_content

0 A fantasy adventure that fuses Greek mythology...
1 Uma Thurman as Medusa, the gorgon with a coiff...
2 With a top-notch cast and dazzling special eff...
3 Whether audiences will get behind The Lightnin...
4 What's really lacking in The Lightning Thief i...
5 It's more a list of ingredients than a movie-m...
6 Harry Potter knockoffs don't come more transpa...
7 Percy Jackson isn't a great movie, but it's a ...
8 Fun, brisk and imaginative
9 Crammed with dragons, set-destroying fights an...
10 This action-packed fantasy adventure, based on...
11 For what it is and for whom it is intended, it...
12 Chris Columbus returns to his comfort zone for...
13 The best thing you can say about Chris Columbu...
14 This cast is simply too generic. None of the y...
15 Percy Jackson may not be "Harry Potter good," ...
16 [P]erfectly inoffensive, occasionally clever...
17 Admirably, the movie isn't bogged down in the ...
18 The main thing this movie seems to be missing ...

19 When the movie slows down to catch its breath,...
 20 Imperfect, yes, but also both suitably imagina...
 21 Although the standard allegorical bases for my...
 22 A little bit worse than lifeless; it's clueles...
 23 You don't even have to be familiar with the fi...
 24 This is generic filmmaking at its most banal, ...
 25 Kids may well enjoy the adventures of The Ligh...
 26 An attempt to steal some of Harry Potter's thu...
 27 With great effects (and a few surprises) it's ...
 28 Columbus directs with his usual frying pan to ...
 29 While this might not rival the later, better H...
 30 Percy Jackson & the Olympians sounds like an O...
 31 The movie feels like a cut-rate Potter knockof...
 32 In Columbus' hands, it once again all breaks d...
 33 The Lightning Thief makes the idea of a sequel...
 34 Columbus aims at nothing more than providing a...
 35 gets so many things right that when it goes as...
 36 The Lightning Thief isn't electric, but it gen...
 37 Chris Columbus and his crew take a literal app...
 38 Even with all its computer-generated fireballs...
 39 The disappearance of lightning simply is not i...
 40 Uma Thurman demonstrates she can wear a snake-...
 41 The Greek-gods premise lets the filmmakers ind...
 42 Of all the players, Uma Thurman channels the B...
 43 Enjoyable, nicely paced family adventure with ...
 44 Action movies of this scale often start off st...
 45 The Lightning Thief is all crass ineptitude an...
 46 The plot is a pretty blatant rehash of the typ...
 47 Young viewers probably will be entertained but...
 48 It raises issues it doesn't have the balls to ...
 49 Whew boy.
 50 My problems with it are the same as with most ...
 51 Sadly worthy of its dumping into the cinematic...
 52 The camera work is frequently pedestrian, the ...
 53 Director Chris Columbus has brought together t...
 54 Dutifully Potteresque it begins, and dutifully...
 55 Has all the CGI sorcery of a Harry Potter pic,...
 56 Columbus knows his way around this kind of mat...
 57 Percy Jackson, you are no Harry Potter. You ar...
 58 Too scary for younger fans of the popular book.
 59 The pleasant surprise is how deftly the story ...

```

[14]: # Dimensions of the critic_reviews DataFrame.
      critic_reviews_df.shape
  
```

```

[14]: (1130017, 8)
  
```

```
[15]: critic_reviews_df.dtypes
```

```
[15]: rotten_tomatoes_link    object
      critic_name            object
      top_critic             bool
      publisher_name         object
      review_type            object
      review_score           object
      review_date            object
      review_content         object
      dtype: object
```

```
[16]: # Display the column names of critic_reviews_df.
      critic_reviews_df.columns
```

```
[16]: Index(['rotten_tomatoes_link', 'critic_name', 'top_critic', 'publisher_name',
          'review_type', 'review_score', 'review_date', 'review_content'],
          dtype='object')
```

3.1 1. Drop Features.

```
[17]: # Dropping the unnecessary features from the critic_reviews_df
      critics_features_to_drop = ["top_critic", "publisher_name",
      ↪ "review_score", "review_content"]
      critic_reviews_df_cleaned = critic_reviews_df.drop(columns =
      ↪ critics_features_to_drop)
```

```
[18]: critic_reviews_df_cleaned.head(60)
```

```
[18]: rotten_tomatoes_link    critic_name review_type review_date
0      m/0814255          Andrew L. Urban    Fresh  2010-02-06
1      m/0814255          Louise Keller      Fresh  2010-02-06
2      m/0814255                NaN      Fresh  2010-02-09
3      m/0814255          Ben McEachen      Fresh  2010-02-09
4      m/0814255          Ethan Alter      Rotten  2010-02-10
5      m/0814255          David Germain      Rotten  2010-02-10
6      m/0814255          Nick Schager      Rotten  2010-02-10
7      m/0814255    Bill Goodykoontz      Fresh  2010-02-10
8      m/0814255          Jordan Hoffman      Fresh  2010-02-10
9      m/0814255          Jim Schembri      Fresh  2010-02-10
10     m/0814255          Mark Adams      Fresh  2010-02-10
11     m/0814255          Roger Moore      Rotten  2010-02-10
12     m/0814255          David Jenkins      Rotten  2010-02-10
13     m/0814255          Joshua Tyler      Fresh  2010-02-11
14     m/0814255          Peter Paras      Rotten  2010-02-11
15     m/0814255          Vic Holtzman      Fresh  2010-02-11
16     m/0814255    MaryAnn Johanson      Fresh  2010-02-11
```

17	m/0814255	Mark Dujsik	Rotten	2010-02-11
18	m/0814255	Nell Minow	Fresh	2010-02-11
19	m/0814255	Mary Elizabeth Williams	Rotten	2010-02-11
20	m/0814255	Dustin Putman	Fresh	2010-02-11
21	m/0814255	Stephen Holden	Rotten	2010-02-11
22	m/0814255	Jeffrey M. Anderson	Rotten	2010-02-11
23	m/0814255	Elizabeth Weitzman	Fresh	2010-02-11
24	m/0814255	Kenneth Turan	Rotten	2010-02-11
25	m/0814255	Moir MacDonald	Rotten	2010-02-11
26	m/0814255	James Rocchi	Fresh	2010-02-11
27	m/0814255	Stephen Whitty	Fresh	2010-02-11
28	m/0814255	David Plotz	Rotten	2010-02-11
29	m/0814255	Jeff Vice	Fresh	2010-02-11
30	m/0814255	Wesley Morris	Rotten	2010-02-11
31	m/0814255	Andrea Gronvall	Rotten	2010-02-11
32	m/0814255	Tasha Robinson	Rotten	2010-02-11
33	m/0814255	Rene Rodriguez	Fresh	2010-02-11
34	m/0814255	Kent Turner	Fresh	2010-02-11
35	m/0814255	Bill Gibron	Rotten	2010-02-11
36	m/0814255	Joe Williams	Fresh	2010-02-11
37	m/0814255	Cole Smithey	Fresh	2010-02-11
38	m/0814255	Michael Phillips	Rotten	2010-02-11
39	m/0814255	Matt Pais	Rotten	2010-02-11
40	m/0814255	Roger Ebert	Fresh	2010-02-11
41	m/0814255	Rich Cline	Fresh	2010-02-11
42	m/0814255	Fionnuala Halligan	Fresh	2010-02-11
43	m/0814255	Matthew Turner	Fresh	2010-02-11
44	m/0814255	Peter Debruge	Fresh	2010-02-11
45	m/0814255	Diva Velez	Rotten	2010-02-12
46	m/0814255	Christopher Smith	Rotten	2010-02-12
47	m/0814255	Brian Tallerico	Rotten	2010-02-12
48	m/0814255	Walter Chaw	Rotten	2010-02-12
49	m/0814255	Nick Starkey	Rotten	2010-02-12
50	m/0814255	Jenna Busch	Rotten	2010-02-12
51	m/0814255	Kurt Loder	Rotten	2010-02-12
52	m/0814255	David Frese	Rotten	2010-02-12
53	m/0814255	Amy Biancolli	Fresh	2010-02-12
54	m/0814255	Rick Groen	Rotten	2010-02-12
55	m/0814255	Adam Markovitz	Rotten	2010-02-12
56	m/0814255	Pete Hammond	Rotten	2010-02-12
57	m/0814255	Tom Long	Rotten	2010-02-12
58	m/0814255	Carrie R. Wheadon	Rotten	2010-02-12
59	m/0814255	Andrea Chase	Fresh	2010-02-12

Explanation for “top_critic”, “publisher_name”, “review_score”, “review_content” dropped: These columns don’t directly influence the prediction of critic scores, and therefore don’t provide predictive value for goals. Narrowing this features reduce the noise and make the

model more focused.

3.2 2. Perform any data extraction/selection steps.

```
[19]: # Count the number of missing (NaN) values in the columns of the DataFrame.
nan_counts = critic_reviews_df_cleaned[['critic_name', 'review_type',
    ↪ 'review_date']].isnull().sum()
print(nan_counts)
```

```
critic_name    18529
review_type      0
review_date      0
dtype: int64
```

```
[20]: # Get unique values in the review_type column.
print(critic_reviews_df_cleaned["review_type"].unique())
```

```
['Fresh' 'Rotten']
```

3.3 3. Transform features if necessary.

```
[21]: # Convert review_date to datetime format.
critic_reviews_df_cleaned["review_date"] = pd.
    ↪ to_datetime(critic_reviews_df_cleaned["review_date"], errors = "coerce")
```

3.4 4. Engineer new useful features.

```
[ ]:
```

3.5 5. Deal with missing data (do not just drop rows or columns without justifying this).

```
[22]: # Replace NaN values in the 'critic_name' column with 'Unknown'
# critic_reviews_df_cleaned['critic_name'].fillna('Unknown', inplace=True)

critic_reviews_df_cleaned['critic_name'] =
    ↪ (critic_reviews_df_cleaned['critic_name'].fillna('Unknown'))
```

3.6 6. Create dummy variables if necessary.

```
[23]: # Create dummy variables for the 'review_type' column.
dummies = pd.get_dummies(critic_reviews_df_cleaned['review_type'],
    ↪ prefix='review_type', dtype = int)

# Concatenate the dummy variables to the original DataFrame.
critic_reviews_df_cleaned = pd.concat([critic_reviews_df_cleaned, dummies],
    ↪ axis=1)
```

```
# Optionally, drop the original 'review_type' column if no longer needed.
critic_reviews_df_cleaned.drop(columns=['review_type'], inplace=True)
```

```
[24]: critic_reviews_df_cleaned.shape
```

```
[24]: (1130017, 5)
```

```
[25]: critic_reviews_df_cleaned.dtypes
```

```
[25]: rotten_tomatoes_link      object
      critic_name              object
      review_date             datetime64[ns]
      review_type_Fresh        int32
      review_type_Rotten        int32
      dtype: object
```

```
[26]: critic_reviews_df_cleaned.head(60)
```

```
[26]: rotten_tomatoes_link      critic_name review_date \
0      m/0814255      Andrew L. Urban  2010-02-06
1      m/0814255      Louise Keller  2010-02-06
2      m/0814255      Unknown  2010-02-09
3      m/0814255      Ben McEachen  2010-02-09
4      m/0814255      Ethan Alter  2010-02-10
5      m/0814255      David Germain  2010-02-10
6      m/0814255      Nick Schager  2010-02-10
7      m/0814255      Bill Goodykoontz  2010-02-10
8      m/0814255      Jordan Hoffman  2010-02-10
9      m/0814255      Jim Schembri  2010-02-10
10     m/0814255      Mark Adams  2010-02-10
11     m/0814255      Roger Moore  2010-02-10
12     m/0814255      David Jenkins  2010-02-10
13     m/0814255      Joshua Tyler  2010-02-11
14     m/0814255      Peter Paras  2010-02-11
15     m/0814255      Vic Holtreman  2010-02-11
16     m/0814255      MaryAnn Johanson  2010-02-11
17     m/0814255      Mark Dujsik  2010-02-11
18     m/0814255      Nell Minow  2010-02-11
19     m/0814255      Mary Elizabeth Williams  2010-02-11
20     m/0814255      Dustin Putman  2010-02-11
21     m/0814255      Stephen Holden  2010-02-11
22     m/0814255      Jeffrey M. Anderson  2010-02-11
23     m/0814255      Elizabeth Weitzman  2010-02-11
24     m/0814255      Kenneth Turan  2010-02-11
25     m/0814255      Moira MacDonald  2010-02-11
26     m/0814255      James Rocchi  2010-02-11
27     m/0814255      Stephen Whitty  2010-02-11
```

28	m/0814255	David Plotz	2010-02-11
29	m/0814255	Jeff Vice	2010-02-11
30	m/0814255	Wesley Morris	2010-02-11
31	m/0814255	Andrea Gronvall	2010-02-11
32	m/0814255	Tasha Robinson	2010-02-11
33	m/0814255	Rene Rodriguez	2010-02-11
34	m/0814255	Kent Turner	2010-02-11
35	m/0814255	Bill Gibron	2010-02-11
36	m/0814255	Joe Williams	2010-02-11
37	m/0814255	Cole Smithey	2010-02-11
38	m/0814255	Michael Phillips	2010-02-11
39	m/0814255	Matt Pais	2010-02-11
40	m/0814255	Roger Ebert	2010-02-11
41	m/0814255	Rich Cline	2010-02-11
42	m/0814255	Fionnuala Halligan	2010-02-11
43	m/0814255	Matthew Turner	2010-02-11
44	m/0814255	Peter Debruge	2010-02-11
45	m/0814255	Diva Velez	2010-02-12
46	m/0814255	Christopher Smith	2010-02-12
47	m/0814255	Brian Tallerico	2010-02-12
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49	m/0814255	Nick Starkey	2010-02-12
50	m/0814255	Jenna Busch	2010-02-12
51	m/0814255	Kurt Loder	2010-02-12
52	m/0814255	David Frese	2010-02-12
53	m/0814255	Amy Biancolli	2010-02-12
54	m/0814255	Rick Groen	2010-02-12
55	m/0814255	Adam Markovitz	2010-02-12
56	m/0814255	Pete Hammond	2010-02-12
57	m/0814255	Tom Long	2010-02-12
58	m/0814255	Carrie R. Wheadon	2010-02-12
59	m/0814255	Andrea Chase	2010-02-12

	review_type_Fresh	review_type_Rotten
0	1	0
1	1	0
2	1	0
3	1	0
4	0	1
5	0	1
6	0	1
7	1	0
8	1	0
9	1	0
10	1	0
11	0	1
12	0	1

13	1	0
14	0	1
15	1	0
16	1	0
17	0	1
18	1	0
19	0	1
20	1	0
21	0	1
22	0	1
23	1	0
24	0	1
25	0	1
26	1	0
27	1	0
28	0	1
29	1	0
30	0	1
31	0	1
32	0	1
33	1	0
34	1	0
35	0	1
36	1	0
37	1	0
38	0	1
39	0	1
40	1	0
41	1	0
42	1	0
43	1	0
44	1	0
45	0	1
46	0	1
47	0	1
48	0	1
49	0	1
50	0	1
51	0	1
52	0	1
53	1	0
54	0	1
55	0	1
56	0	1
57	0	1
58	0	1
59	1	0

4 MOVIES

```
[27]: # Display the first few rows of movies_df to understand structure.
      movies_df.head(60)
```

```
[27]:          rotten_tomatoes_link  \
0                               m/0814255
1                               m/0878835
2                               m/10
3                               m/1000013-12_angry_men
4       m/1000079-20000_leagues_under_the_sea
5                               m/10000_bc
6                               m/1000121-39_steps
7                               m/1000123-310_to_yuma
8                               m/10002008-charly
9       m/1000204-abraham_lincoln
10          m/10002114-dark_water
11          m/1000224-accused
12          m/10002516-lost_city
13          m/10002519-breaking_point
14          m/1000253-adams_rib
15          m/10002635-bridge_of_san_luis_rey
16          m/10002673-prowler
17          m/10003276-criminal
18          m/1000334-adventures_of_mark_twain
19          m/10003437-deep_blue
20          m/1000355-adventures_of_robin_hood
21          m/10003876-man_hunt
22          m/10003925-dead_end
23          m/10004050-bandwagon
24          m/10004069-whore
25          m/10004209-tristan_and_isolde
26          m/10004226-wild_side
27          m/10004232-10004232-tale_of_two_pizzas
28          m/10004288-running_scared
29          m/10004352-going_the_distance
30          m/10004504-ultraviolet
31          m/10004635-home_of_the_brave
32          m/10004659-arthur
33          m/10004684-malevolence
34          m/10004697-eternal
35          m/10004835-farewell_to_arms
36          m/10004890-narrow_margin
37          m/10004925-matador
38          m/10005057-smile
39          m/10005104-intruder
40          m/10005178-saint_ralph
```

41 m/10005403-the_baxter
 42 m/10005499-oliver_twist
 43 m/1000560-alamo
 44 m/10005755-g
 45 m/1000580-alexander_the_great
 46 m/1000581-alfie
 47 m/10005819-10005819-neverwas
 48 m/10006007-the_second_chance
 49 m/10006057-moonlight
 50 m/1000617-aliens
 51 m/10006188-rise
 52 m/1000626-all_about_eve
 53 m/10006270
 54 m/10006324
 55 m/10006343-dirty
 56 m/10006370-stay_alive
 57 m/1000640-all_of_me
 58 m/1000643-all_quiet_on_the_western_front
 59 m/10006434-dreamland

	movie_title \
0	Percy Jackson & the Olympians: The Lightning T...
1	Please Give
2	10
3	12 Angry Men (Twelve Angry Men)
4	20,000 Leagues Under The Sea
5	10,000 B.C.
6	The 39 Steps
7	3:10 to Yuma
8	Charly (A Heartbeat Away)
9	Abraham Lincoln
10	Dark Water
11	The Accused
12	The Lost City
13	The Breaking Point
14	Adam's Rib
15	The Bridge of San Luis Rey
16	The Prowler (Cost of Living)
17	Criminal
18	The Adventures of Mark Twain
19	Deep Blue
20	The Adventures of Robin Hood
21	Man Hunt
22	Dead End
23	Bandwagon
24	Whore
25	Tristan & Isolde

26 Wild Side
 27 A Tale of Two Pizzas
 28 Running Scared
 29 Going the Distance
 30 Ultraviolet
 31 Home of the Brave
 32 Arthur and the Invisibles (Arthur and the Mini...
 33 Malevolence
 34 Eternal
 35 A Farewell To Arms
 36 The Narrow Margin
 37 The Matador
 38 Smile
 39 The Intruder (L'Intrus)
 40 Saint Ralph
 41 The Baxter
 42 Oliver Twist
 43 The Alamo
 44 G
 45 Alexander the Great
 46 Alfie
 47 Neverwas
 48 The Second Chance
 49 Moonlight
 50 Aliens
 51 Rise: Blood Hunter
 52 All About Eve
 53 When Do We Eat?
 54 The Mostly Unfabulous Social Life of Ethan Green
 55 Dirty
 56 Stay Alive
 57 All of Me
 58 All Quiet on the Western Front
 59 Dreamland

movie_info \

0 Always trouble-prone, the life of teenager Per...
 1 Kate (Catherine Keener) and her husband Alex (...
 2 A successful, middle-aged Hollywood songwriter...
 3 Following the closing arguments in a murder tr...
 4 In 1866, Professor Pierre M. Aronnax (Paul Luk...
 5 Mammoth hunter D'Leh (Steven Strait) has long ...
 6 While on vacation in London, Canadian Richard ...
 7 Dan Evans (Van Heflin), a drought-plagued Ariz...
 8 Cultural differences, past loves and personal ...
 9 The 16th U.S. president (Walter Huston) is por...
 10 In this moody Japanese horror film, newly-sing...

11 Out drinking one night after a fight with her ...
 12 Fico Fellove (Andy Garcia), an apolitical Hava...
 13 A charter-boat captain winds up in the middle ...
 14 A courtroom rivalry finds its way into the hou...
 15 During the Spanish Inquisition, Franciscan mon...
 16 After being frightened by a peeping Tom at her...
 17 Needing a new partner capable of intricate con...
 18 In this claymation film, celebrated author Mar...
 19 This nature documentary shows the lives of dif...
 20 When King Richard the Lionheart is captured, h...
 21 Renowned big-game hunter Captain Thorndike (Wa...
 22 When a family en route to a Christmas Eve gath...
 23 When Tony (Lee Holmes) loses his job, he decid...
 24 Encouraged by her neighbor (Daryl Hannah), a f...
 25 After the fall of the Roman Empire, English or...
 26 Along with one (Edouard Nikitine) of her live-...
 27 The offspring of rival pizzeria owners (Vincen...
 28 Mafia flunky Joey Gazelle (Paul Walker) is ord...
 29 Nick (Christopher Jacot) is a successful young...
 30 In the late 21st-century, a vampiric disease m...
 31 In 1965, Klansmen murder a civil-rights worker...
 32 Ten-year-old Arthur must find a way to keep th...
 33 Bank robbers (Brandon Johnson, Heather Magee) ...
 34 A detective's (Conrad Pla) search for his miss...
 35 Based on the Ernest Hemingway novel, this Worl...
 36 Tense story of a tough cop attempting to trans...
 37 Salesman Danny Wright (Greg Kinnear) is workin...
 38 While visiting China, an American high-school ...
 39 After receiving a heart transplant, an enigmat...
 40 The year 1954 finds teen outcast Ralph Walker ...
 41 Elliot Sherman (Michael Showalter) has had ter...
 42 An orphan (Barney Clark) in 19th-century Londo...
 43 In 1836, the Mexican army, led by General Sant...
 44 Summer G (Richard T. Jones), an aspiring rappe...
 45 The son of Philip of Macedonia (Fredric March)...
 46 Set in postwar London, "Alfie" features Michae...
 47 Dr. Zach Riley (Aaron Eckhart) begins practici...
 48 Two pastors, one from a wealthy parish in a pr...
 49 Claire, a Dutch girl, discovers a wounded boy ...
 50 After floating in space for 57 years, Lt. Rippl...
 51 Sadie (Lucy Liu), a reporter, investigates an ...
 52 Backstage story revolving around aspiring actr...
 53 Members of the Stuckman family gather for Sede...
 54 A young man named Ethan (Daniel Letterle) is h...
 55 Armando Sancho (Clifton Collins Jr.) is a form...
 56 Teenagers decide to pass idle time by playing ...
 57 While on her deathbed, the rich Edwina Cutwate...

58 At the start of World War I, Paul Baumer (Rich...
 59 Teenage Aubrey (Agnes Bruckner) is always ther...

	critics_consensus	content_rating	\
0	Though it may seem like just another Harry Pot...	PG	
1	Nicole Holofcener's newest might seem slight i...	R	
2	Blake Edwards' bawdy comedy may not score a pe...	R	
3	Sidney Lumet's feature debut is a superbly wri...	NR	
4	One of Disney's finest live-action adventures,...	G	
5	With attention strictly paid to style instead ...	PG-13	
6	Packed with twists and turns, this essential e...	NR	
7	NaN	NR	
8	NaN	PG	
9	NaN	NR	
10	NaN	PG-13	
11	NaN	R	
12	Its heart is in the right place, but what star...	R	
13	NaN	NR	
14	Matched by Garson Kanin's witty, sophisticated...	NR	
15	Despite an all-star cast and some impressive v...	PG	
16	NaN	PG	
17	If you saw Nine Queens, it may feel redundant,...	R	
18	NaN	G	
19	Full of visually striking footage, Deep Blue m...	G	
20	Errol Flynn thrills as the legendary title cha...	PG	
21	NaN	NR	
22	NaN	R	
23	NaN	R	
24	NaN	R	
25	Competent but somewhat static, Tristan & Isold...	PG-13	
26	NaN	NR	
27	NaN	PG	
28	This film runs with frenetic energy punctuated...	R	
29	NaN	R	
30	An incomprehensible and forgettable sci-fi thr...	PG-13	
31	NaN	R	
32	Arthur wastes its big-name voice talent on a p...	PG	
33	NaN	R	
34	Even by the standards of camp, Eternal fails t...	R	
35	NaN	NR	
36	NaN	NR	
37	This humorously amoral, oddball comic thriller...	R	
38	NaN	PG-13	
39	The impressionistic narrative may confound the...	G	
40	Saint Ralph is an amiable tear-jerker that wal...	PG-13	
41	The Baxter is good-natured, but there are simp...	PG-13	
42	Polanski's version of Dickens' classic won't h...	PG-13	

43	John Wayne proves to be less compelling behind...	NR
44	Race and class struggles come to the Hamptons ...	R
45	NaN	G
46	Anchored by Michael Caine's charmingly droll p...	PG
47	NaN	PG-13
48	NaN	PG-13
49	NaN	R
50	While Alien was a marvel of slow-building, atm...	R
51	NaN	R
52	Smart, sophisticated, and devastatingly funny,...	PG
53	A crass, sitcom-ish holiday comedy, When Do We...	R
54	Ethan Green makes for an annoying protagonist,...	R
55	Dirty is an unoriginal L.A. cop drama that wea...	R
56	A by-the-numbers teen horror flick, Stay Alive...	PG-13
57	A high-concept farce carried by Carl Reiner's ...	PG
58	NaN	NR
59	NaN	PG-13

	genres \
0	Action & Adventure, Comedy, Drama, Science Fic...
1	Comedy
2	Comedy, Romance
3	Classics, Drama
4	Action & Adventure, Drama, Kids & Family
5	Action & Adventure, Classics, Drama
6	Action & Adventure, Classics, Mystery & Suspense
7	Classics, Drama, Western
8	Comedy, Drama, Romance
9	Classics, Drama
10	Art House & International, Horror, Mystery & S...
11	Drama, Mystery & Suspense
12	Drama
13	Drama
14	Classics, Comedy, Romance
15	Art House & International, Drama, Faith & Spir...
16	Drama, Mystery & Suspense
17	Drama, Mystery & Suspense
18	Action & Adventure, Animation, Kids & Family
19	Documentary, Special Interest
20	Action & Adventure, Classics, Romance
21	Classics, Drama, Mystery & Suspense
22	Comedy, Horror, Mystery & Suspense
23	Comedy
24	Art House & International, Drama
25	Drama, Science Fiction & Fantasy, Romance
26	Art House & International, Drama
27	Comedy

28 Action & Adventure, Drama, Mystery & Suspense
 29 Comedy
 30 Action & Adventure, Drama, Horror, Science Fic...
 31 Action & Adventure, Documentary, Drama, Specia...
 32 Animation, Kids & Family, Science Fiction & Fa...
 33 Horror
 34 Horror, Mystery & Suspense
 35 Classics, Drama, Romance
 36 Classics, Drama, Mystery & Suspense
 37 Action & Adventure, Comedy, Drama, Mystery & S...
 38 Art House & International, Comedy, Drama
 39 Art House & International, Drama
 40 Comedy, Drama
 41 Comedy, Romance
 42 Drama, Kids & Family
 43 Action & Adventure, Classics, Drama, Western
 44 Drama, Romance
 45 Classics
 46 Classics, Comedy, Drama, Romance
 47 Drama, Mystery & Suspense
 48 Drama, Faith & Spirituality
 49 Drama, Mystery & Suspense
 50 Action & Adventure, Horror, Science Fiction & ...
 51 Horror, Mystery & Suspense
 52 Classics, Drama
 53 Comedy
 54 Comedy, Drama
 55 Drama
 56 Horror, Mystery & Suspense
 57 Comedy, Science Fiction & Fantasy
 58 Drama
 59 Art House & International, Drama

directors \
 0 Chris Columbus
 1 Nicole Holofcener
 2 Blake Edwards
 3 Sidney Lumet
 4 Richard Fleischer
 5 Roland Emmerich
 6 Alfred Hitchcock
 7 Delmer Daves
 8 Adam Thomas Anderegg
 9 D.W. Griffith
 10 Hideo Nakata
 11 Jonathan Kaplan
 12 Andy Garcia

13	Michael Curtiz
14	George Cukor
15	Mary McGuckian
16	Joseph Losey
17	Greggory Jacobs, Gregory Jacobs, Greg Jacobs
18	Will Vinton
19	Andy Byatt, Alastair Fothergill
20	Michael Curtiz, William Keighley
21	Fritz Lang
22	Jean-Baptiste Andrea, Fabrice Canepa, Jean-Bap...
23	John Schultz
24	MarÃa LidÃ³n
25	Kevin Reynolds
26	SÃbastien Lifshitz
27	Vincent Sassone
28	Wayne Kramer
29	Mark Griffiths
30	Kurt Wimmer
31	Paola di Florio
32	Luc Besson
33	Stevan Mena
34	Wilhelm Liebenberg, Federico Sanchez
35	Charles Vidor, John Hudson, John Huston
36	Richard Fleischer
37	Bob Yari, Richard Shepard
38	Jeff Kramer, Jeffrey Kramer
39	Claire Denis
40	Michael McGowan
41	Michael Showalter
42	Roman Polanski
43	John Wayne
44	Christopher Scott Cherot
45	Robert Rossen
46	Lewis Gilbert
47	Joshua Michael Stern
48	Steve Taylor
49	Paula Van der Oest
50	James Cameron
51	Sebastian Gutierrez
52	Joseph L. Mankiewicz
53	Salvador Litvak
54	George Bamber
55	Chris Fisher
56	William Brent Bell
57	Carl Reiner
58	Delbert Mann
59	Jason Matzner

0 authors \
 1 Craig Titley, Chris Columbus, Rick Riordan
 2 Nicole Holofcener
 3 Blake Edwards
 4 Reginald Rose
 5 Earl Felton
 6 Harald Kloser, Roland Emmerich
 7 Alma Reville, Charles Bennett, Ian Hay, Alfred...
 8 Halsted Welles
 9 Jack Weyland, Janine Whetton Gilbert
 10 Gerrit J. Lloyd, Stephen Vincent Ben  t
 11 Hideo Hakata, Takashige Ichise, Yoshihiro Naka...
 12 Tom Topor
 13 G. Cabrera Infante, Guillermo Cabrera Infante
 14 Ranald MacDougall
 15 Garson Kanin, Ruth Gordon
 16 Mary McGuckian
 17 Robert Thoeren, Hugo Butler, Dalton Trumbo
 18 Gregory Jacobs, Fabi  n Bielinsky, Steven Sode...
 19 Susan Shadburne
 20 Alastair Fothergill, Andy Byatt, Tim Ecott
 21 Norman Reilly Raine, Seton I. Miller, Michael ...
 22 Dudley Nichols
 23 Jean-Baptiste Andrea, Fabrice Canepa
 24 John Schultz
 25 Adela Iba  ez, Isabel Pisano
 26 Dean Georgaris
 27 S  bastien Lifshitz, St  phane Bouquet
 28 Vincent Sassone
 29 Wayne Kramer
 30 Eric Goodman, Kelly Senecal
 31 Kurt Wimmer
 32 Paola di Florio
 33 Luc Besson, Celine Garcia
 34 Stevan Mena
 35 Wilhelm Liebenberg, Federico Sanchez
 36 Ben Hecht
 37 Earl Felton, Martin Goldsmith
 38 Richard Shepard, Richard Shepherd
 39 Jeff Kramer, Jeffrey Kramer
 40 Claire Denis, Jean-Pol Fargeau
 41 Michael McGowan
 42 Michael Showalter
 43 Ronald Harwood
 44 James Edward Grant
 45 Charles E. Drew, Christopher Scott Cherot, Cha...

45 Robert Rossen
 46 Bill Naughton
 47 Joshua Michael Stern
 48 Steve Taylor, Chip Arnold, Ben Pearson, Henry ...
 49 Carel Donck
 50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak, Nina Davidovich
 54 David Vernon
 55 Chris Fisher, Gil Reavill, Eric Saks
 56 William Brent Bell, Matthew Peterman
 57 Henry Olek, Carl Reiner, Phil Alden Robinson
 58 Paul Monash
 59 Tom Willett, Jason Matzner

	actors	original_release_date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01
5	Steven Strait, Camilla Belle, Cliff Curtis, Jo...	2008-03-07
6	Robert Donat, Madeleine Carroll, Godfrey Tearl...	1935-08-01
7	Glenn Ford, Van Heflin, Felicia Farr, Leora Da...	1957-08-07
8	Heather Beers, Gary Neilson, Lisa McCammon, Ja...	2002-09-27
9	Walter Huston, Una Merkel, Kay Hammond, Ian Ke...	1930-11-08
10	Hitomi Kuroki, Rio Kanno, Shigemitsu Ogi, Mire...	2002-01-19
11	Jodie Foster, Kelly McGillis, Bernie Coulson, ...	1988-10-14
12	Andy Garcia, Dustin Hoffman, Bill Murray, InÃ©...	2005-09-03
13	John Garfield, Patricia Neal, Phyllis Thaxter,...	1950-10-06
14	Spencer Tracy, Katharine Hepburn, Judy Hollida...	1949-11-18
15	Gabriel Byrne, F. Murray Abraham, Kathy Bates,...	2005-06-10
16	Van Heflin, Evelyn Keyes, Katherine Warren, Jo...	1951-12-03
17	John C. Reilly, Diego Luna, Maggie Gyllenhaal,...	2004-09-24
18	James Whitmore, Michele Mariana, Gary Krug, Ch...	1985-03-01
19	Pierce Brosnan, Michael Gambon	2005-06-17
20	Errol Flynn, Olivia de Havilland, Basil Rathbo...	1938-05-14
21	Walter Pidgeon, Joan Bennett, George Sanders, ...	1941-01-01
22	Ray Wise, Lin Shaye, Mick Cain, Alexandra Hold...	2003-09-26
23	Matthew Hennessey, Kevin Corrigan, Steve Parla...	1997-09-12
24	Daryl Hannah, Denise Richards, Joaquim de Alme...	2004-04-16
25	James Franco, Sophia Myles, Rufus Sewell, Davi...	2006-01-13
26	StÃ©phanie Michelini, Josiane Stoleru, Yasmine...	2005-06-10
27	Vincent Pastore, Vincent Sassone, Frank Vincen...	2003-11-07
28	Paul Walker, Cameron Bright, Vera Farmiga, Kar...	2006-02-24
29	Christopher Jacot, Shawn Roberts, Ryan Bellevi...	2004-08-20

30	Milla Jovovich, Cameron Bright, Nick Chinlund,...	2006-03-03
31	Mary Liuzzo, Tony Liuzzo, Sally Liuzzo, Penny ...	2004-05-11
32	Freddie Highmore, Mia Farrow, Penny Balfour, D...	2007-01-12
33	Samantha Dark, R. Brandon Johnson, Heather Mag...	2004-09-10
34	Sarah Manninen, Caroline Neron, Victoria Sanch...	2004-08-26
35	Rock Hudson, Jennifer Jones, Vittorio De Sica,...	1957-12-14
36	Charlie MacGraw, Charles McGraw, Marie Windsor...	1952-05-04
37	Pierce Brosnan, Greg Kinnear, Hope Davis, Phil...	2006-01-27
38	Mika Boorem, Wang Luoyong, Yi Ding (II), Beau ...	2005-04-08
39	Beatrice Dalle, Michael Subor, Jean-Marc Terii...	2005-12-23
40	Adam Butcher, Campbell Scott, Gordon Pinsent, ...	2005-08-05
41	Michael Showalter, Elizabeth Banks, Michelle W...	2005-08-26
42	Ben Kingsley, Barney Clark, Leanne Rowe, Mark ...	2005-09-30
43	John Wayne, Richard Widmark, Laurence Harvey, ...	1960-10-24
44	Richard T. Jones, Blair Underwood, Chenoa Maxw...	2005-09-16
45	Richard Burton, Fredric March, Claire Bloom, D...	1956-03-28
46	Michael Caine, Millicent Martin, Julia Foster,...	1966-08-24
47	Aaron Eckhart, Ian McKellen, Brittany Murphy, ...	2005-09-09
48	Michael W. Smith, Jeff Obafemi Carr, Jonathan ...	2006-02-17
49	Laurien Van Den Broeck, Hunter Bussemaker, Joh...	2003-10-17
50	Sigourney Weaver, Carrie Henn, Michael Biehn, ...	1986-07-18
51	Lucy Liu, Michael Chiklis, Carla Gugino, James...	2007-06-01
52	Bette Davis, Anne Baxter, Gary Merrill, George...	1950-01-01
53	Michael Lerner, Lesley Ann Warren, Jack Klugma...	2006-04-07
54	Daniel Letterle, Meredith Baxter, David Monaha...	2006-06-16
55	Cuba Gooding Jr., Wyclef Jean, Keith David, Cl...	2006-02-24
56	Jon Foster, Frankie Muniz, Samaire Armstrong, ...	2006-03-24
57	Steve Martin, Lily Tomlin, Victoria Tennant, M...	1984-09-21
58	Richard Thomas, Ernest Borgnine, Ian Holm, Don...	NaT
59	Agnes Bruckner, Kelli Garner, Justin Long, Gin...	2006-12-19

	production_company	tomatometer_status	tomatometer_rating \
0	20th Century Fox	Rotten	49.0
1	Sony Pictures Classics	Certified-Fresh	87.0
2	Waner Bros.	Fresh	67.0
3	Criterion Collection	Certified-Fresh	100.0
4	Disney	Fresh	89.0
5	Warner Bros. Pictures	Rotten	8.0
6	Gaumont British Distributors	Certified-Fresh	96.0
7	Columbia Pictures	Fresh	96.0
8	Excel Entertainment	Rotten	20.0
9	United Artists	Fresh	82.0
10	Toho Company Ltd	Fresh	80.0
11	Paramount Pictures	Fresh	91.0
12	Magnolia Pictures	Rotten	25.0
13	Warner Home Video	Fresh	100.0
14	MGM Home Entertainment	Fresh	96.0

15	...	Fine Line Features	Rotten	4.0
16	...	VCI Entertainment	Fresh	100.0
17	...	Warner Bros. Pictures	Fresh	69.0
18	...	Eureka Entertainment Ltd	Fresh	80.0
19	...	Miramax Flims	Fresh	67.0
20	...	Warner Bros.	Certified-Fresh	100.0
21	...	Twentieth Century Fox	Fresh	92.0
22	...	LionsGate Entertainment	Fresh	75.0
23	...	Avalanche Entertainment	Fresh	80.0
24	...	Dolores Pictures LLC	Rotten	0.0
25	...	20th Century Fox	Rotten	31.0
26	...	Wellspring	Fresh	63.0
27	...	NaN	Rotten	38.0
28	...	New Line Cinema	Rotten	41.0
29	...	SÃ©ville Pictures	Rotten	0.0
30	...	Sony Pictures	Rotten	8.0
31	...	Emerging Pictures	Fresh	93.0
32	...	MGM	Rotten	22.0
33	...	Anchor Bay Entertainment	Rotten	33.0
34	...	Regent Releasing	Rotten	21.0
35	...	Fox	Rotten	0.0
36	...	Warner Home Video	Fresh	100.0
37	...	The Weinstein Company	Certified-Fresh	75.0
38	...	Dark Forest Releasing	Rotten	15.0
39	...	Wellspring	Fresh	87.0
40	...	IDP	Fresh	64.0
41	...	IFC Films	Rotten	32.0
42	...	Sony Pictures	Fresh	61.0
43	...	Batjac Productions	Rotten	52.0
44	...	Slow Hand Cinema	Rotten	24.0
45	...	United Artists	Rotten	0.0
46	...	Paramount Pictures	Fresh	96.0
47	...	Senator Film	Rotten	14.0
48	...	Sony Pictures	Rotten	44.0
49	...	Indican Pictures	Fresh	60.0
50	...	20th Century Fox	Certified-Fresh	97.0
51	...	IDP Distribution	Rotten	36.0
52	...	20th Century Fox	Certified-Fresh	99.0
53	...	ThinkFilm	Rotten	42.0
54	...	Regent Releasing	Rotten	30.0
55	...	SNP Inc.	Rotten	21.0
56	...	Buena Vista	Rotten	10.0
57	...	HBO Video	Certified-Fresh	85.0
58	...	Unknown	Fresh	100.0
59	...	Dreamland Motion Picture Co.	Fresh	67.0

tomatometer_count audience_status audience_rating audience_count \

0	149.0	Spilled	53.0	254421.0
1	142.0	Upright	64.0	11574.0
2	24.0	Spilled	53.0	14684.0
3	54.0	Upright	97.0	105386.0
4	27.0	Upright	74.0	68918.0
5	149.0	Spilled	37.0	411140.0
6	51.0	Upright	86.0	23890.0
7	28.0	Upright	79.0	9243.0
8	10.0	Upright	87.0	4819.0
9	11.0	Spilled	40.0	457.0
10	15.0	Upright	66.0	21475.0
11	22.0	Upright	79.0	20821.0
12	83.0	Upright	64.0	25944.0
13	10.0	Upright	86.0	335.0
14	28.0	Upright	86.0	10563.0
15	24.0	Spilled	35.0	1935.0
16	18.0	Upright	86.0	286.0
17	124.0	Spilled	57.0	6711.0
18	5.0	Upright	82.0	1046.0
19	52.0	Upright	80.0	4692.0
20	48.0	Upright	89.0	33946.0
21	12.0	Upright	75.0	554.0
22	8.0	Upright	60.0	10373.0
23	5.0	Upright	91.0	386.0
24	6.0	Upright	63.0	914.0
25	121.0	Upright	71.0	220956.0
26	16.0	Upright	80.0	867.0
27	8.0	NaN	NaN	NaN
28	131.0	Upright	79.0	135400.0
29	5.0	Upright	61.0	13033.0
30	83.0	Spilled	30.0	279994.0
31	15.0	Spilled	48.0	866.0
32	92.0	Spilled	56.0	134483.0
33	18.0	Upright	83.0	2622.0
34	34.0	Spilled	39.0	1295.0
35	12.0	Spilled	37.0	1460.0
36	11.0	Upright	88.0	1135.0
37	156.0	Upright	65.0	73701.0
38	13.0	Upright	66.0	2059.0
39	30.0	Upright	73.0	9701.0
40	69.0	Upright	85.0	6203.0
41	75.0	Upright	75.0	7436.0
42	143.0	Upright	63.0	23294.0
43	21.0	Upright	63.0	7337.0
44	29.0	Upright	60.0	1143.0
45	8.0	Spilled	42.0	5383.0
46	27.0	Upright	76.0	14669.0

47	7.0	Upright	65.0	4059.0
48	16.0	Upright	78.0	79542.0
49	10.0	Upright	67.0	1430.0
50	76.0	Upright	94.0	429674.0
51	14.0	Spilled	20.0	8239.0
52	77.0	Upright	94.0	45248.0
53	38.0	Upright	68.0	1121.0
54	27.0	Spilled	58.0	2394.0
55	28.0	Spilled	42.0	7326.0
56	59.0	Spilled	55.0	113424.0
57	41.0	Upright	67.0	14346.0
58	5.0	Upright	69.0	7588.0
59	15.0	Upright	65.0	5650.0

	tomatometer_top_critics_count	tomatometer_fresh_critics_count	\
0	43	73	
1	44	123	
2	2	16	
3	6	54	
4	5	24	
5	37	12	
6	8	49	
7	6	27	
8	0	2	
9	4	9	
10	3	12	
11	1	20	
12	37	21	
13	1	10	
14	6	27	
15	13	1	
16	6	18	
17	35	85	
18	1	4	
19	18	35	
20	8	48	
21	0	11	
22	1	6	
23	1	4	
24	4	0	
25	36	38	
26	4	10	
27	5	3	
28	39	54	
29	2	0	
30	21	7	
31	7	14	

32	30	20
33	4	6
34	10	7
35	6	0
36	0	11
37	39	117
38	6	2
39	10	26
40	26	44
41	25	24
42	39	87
43	4	11
44	17	7
45	1	0
46	6	26
47	1	1
48	8	7
49	6	6
50	17	74
51	2	5
52	22	76
53	19	16
54	13	8
55	10	6
56	18	6
57	10	35
58	0	5
59	5	10

	tomatometer_rotten_critics_count
0	76
1	19
2	8
3	0
4	3
5	137
6	2
7	1
8	8
9	2
10	3
11	2
12	62
13	0
14	1
15	23
16	0

17	39
18	1
19	17
20	0
21	1
22	2
23	1
24	6
25	83
26	6
27	5
28	77
29	5
30	76
31	1
32	72
33	12
34	27
35	12
36	0
37	39
38	11
39	4
40	25
41	51
42	56
43	10
44	22
45	8
46	1
47	6
48	9
49	4
50	2
51	9
52	1
53	22
54	19
55	22
56	53
57	6
58	0
59	5

[60 rows x 22 columns]

```
[28]: # Dimensions of the movies_df DataFrame.  
movies_df.shape
```

```
[28]: (17712, 22)
```

```
[29]: movies_df.dtypes
```

```
[29]: rotten_tomatoes_link      object  
movie_title                  object  
movie_info                   object  
critics_consensus           object  
content_rating               object  
genres                       object  
directors                    object  
authors                      object  
actors                       object  
original_release_date        datetime64[ns]  
streaming_release_date        datetime64[ns]  
runtime                      float64  
production_company           object  
tomatometer_status           object  
tomatometer_rating           float64  
tomatometer_count            float64  
audience_status             object  
audience_rating             float64  
audience_count              float64  
tomatometer_top_critics_count int64  
tomatometer_fresh_critics_count int64  
tomatometer_rotten_critics_count int64  
dtype: object
```

```
[30]: # Display the column names of movies_df.  
movies_df.columns
```

```
[30]: Index(['rotten_tomatoes_link', 'movie_title', 'movie_info',  
        'critics_consensus', 'content_rating', 'genres', 'directors', 'authors',  
        'actors', 'original_release_date', 'streaming_release_date', 'runtime',  
        'production_company', 'tomatometer_status', 'tomatometer_rating',  
        'tomatometer_count', 'audience_status', 'audience_rating',  
        'audience_count', 'tomatometer_top_critics_count',  
        'tomatometer_fresh_critics_count', 'tomatometer_rotten_critics_count'],  
        dtype='object')
```

4.1 1. Drop Features.

```
[31]: # Dropping the unnecessary features from the movies_df.
movies_features_to_drop = ['rotten_tomatoes_link', 'movie_info',
    ↪ 'production_company', 'tomatometer_status', 'audience_status',
    ↪ 'original_release_date']
movies_df_cleaned = movies_df.drop(columns = movies_features_to_drop)
```

```
[32]: movies_df_cleaned.head(60)
```

```
[32]:
      movie_title \
0  Percy Jackson & the Olympians: The Lightning T...
1                Please Give
2                        10
3      12 Angry Men (Twelve Angry Men)
4      20,000 Leagues Under The Sea
5                10,000 B.C.
6                The 39 Steps
7                3:10 to Yuma
8      Charly (A Heartbeat Away)
9      Abraham Lincoln
10             Dark Water
11             The Accused
12             The Lost City
13             The Breaking Point
14             Adam's Rib
15             The Bridge of San Luis Rey
16             The Prowler (Cost of Living )
17             Criminal
18             The Adventures of Mark Twain
19             Deep Blue
20             The Adventures of Robin Hood
21             Man Hunt
22             Dead End
23             Bandwagon
24             Whore
25             Tristan & Isolde
26             Wild Side
27             A Tale of Two Pizzas
28             Running Scared
29             Going the Distance
30             Ultraviolet
31             Home of the Brave
32  Arthur and the Invisibles (Arthur and the Mini...
33             Malevolence
34             Eternal
35             A Farewell To Arms
```

36	The Narrow Margin
37	The Matador
38	Smile
39	The Intruder (L'Intrus)
40	Saint Ralph
41	The Baxter
42	Oliver Twist
43	The Alamo
44	G
45	Alexander the Great
46	Alfie
47	Neverwas
48	The Second Chance
49	Moonlight
50	Aliens
51	Rise: Blood Hunter
52	All About Eve
53	When Do We Eat?
54	The Mostly Unfabulous Social Life of Ethan Green
55	Dirty
56	Stay Alive
57	All of Me
58	All Quiet on the Western Front
59	Dreamland

	critics_consensus	content_rating	\
0	Though it may seem like just another Harry Pot...	PG	
1	Nicole Holofcener's newest might seem slight i...	R	
2	Blake Edwards' bawdy comedy may not score a pe...	R	
3	Sidney Lumet's feature debut is a superbly wri...	NR	
4	One of Disney's finest live-action adventures,...	G	
5	With attention strictly paid to style instead ...	PG-13	
6	Packed with twists and turns, this essential e...	NR	
7	NaN	NR	
8	NaN	PG	
9	NaN	NR	
10	NaN	PG-13	
11	NaN	R	
12	Its heart is in the right place, but what star...	R	
13	NaN	NR	
14	Matched by Garson Kanin's witty, sophisticated...	NR	
15	Despite an all-star cast and some impressive v...	PG	
16	NaN	PG	
17	If you saw Nine Queens, it may feel redundant,...	R	
18	NaN	G	
19	Full of visually striking footage, Deep Blue m...	G	
20	Errol Flynn thrills as the legendary title cha...	PG	

21		NaN	NR
22		NaN	R
23		NaN	R
24		NaN	R
25	Competent but somewhat static, Tristan & Isold...		PG-13
26		NaN	NR
27		NaN	PG
28	This film runs with frenetic energy punctuated...		R
29		NaN	R
30	An incomprehensible and forgettable sci-fi thr...		PG-13
31		NaN	R
32	Arthur wastes its big-name voice talent on a p...		PG
33		NaN	R
34	Even by the standards of camp, Eternal fails t...		R
35		NaN	NR
36		NaN	NR
37	This humorously amoral, oddball comic thriller...		R
38		NaN	PG-13
39	The impressionistic narrative may confound the...		G
40	Saint Ralph is an amiable tear-jerker that wal...		PG-13
41	The Baxter is good-natured, but there are simp...		PG-13
42	Polanski's version of Dickens' classic won't h...		PG-13
43	John Wayne proves to be less compelling behind...		NR
44	Race and class struggles come to the Hamptons ...		R
45		NaN	G
46	Anchored by Michael Caine's charmingly droll p...		PG
47		NaN	PG-13
48		NaN	PG-13
49		NaN	R
50	While Alien was a marvel of slow-building, atm...		R
51		NaN	R
52	Smart, sophisticated, and devastatingly funny,...		PG
53	A crass, sitcom-ish holiday comedy, When Do We...		R
54	Ethan Green makes for an annoying protagonist,...		R
55	Dirty is an unoriginal L.A. cop drama that wea...		R
56	A by-the-numbers teen horror flick, Stay Alive...		PG-13
57	A high-concept farce carried by Carl Reiner's ...		PG
58		NaN	NR
59		NaN	PG-13

	genres \
0	Action & Adventure, Comedy, Drama, Science Fic...
1	Comedy
2	Comedy, Romance
3	Classics, Drama
4	Action & Adventure, Drama, Kids & Family
5	Action & Adventure, Classics, Drama

6 Action & Adventure, Classics, Mystery & Suspense
 7 Classics, Drama, Western
 8 Comedy, Drama, Romance
 9 Classics, Drama
 10 Art House & International, Horror, Mystery & S...
 11 Drama, Mystery & Suspense
 12 Drama
 13 Drama
 14 Classics, Comedy, Romance
 15 Art House & International, Drama, Faith & Spir...
 16 Drama, Mystery & Suspense
 17 Drama, Mystery & Suspense
 18 Action & Adventure, Animation, Kids & Family
 19 Documentary, Special Interest
 20 Action & Adventure, Classics, Romance
 21 Classics, Drama, Mystery & Suspense
 22 Comedy, Horror, Mystery & Suspense
 23 Comedy
 24 Art House & International, Drama
 25 Drama, Science Fiction & Fantasy, Romance
 26 Art House & International, Drama
 27 Comedy
 28 Action & Adventure, Drama, Mystery & Suspense
 29 Comedy
 30 Action & Adventure, Drama, Horror, Science Fic...
 31 Action & Adventure, Documentary, Drama, Specia...
 32 Animation, Kids & Family, Science Fiction & Fa...
 33 Horror
 34 Horror, Mystery & Suspense
 35 Classics, Drama, Romance
 36 Classics, Drama, Mystery & Suspense
 37 Action & Adventure, Comedy, Drama, Mystery & S...
 38 Art House & International, Comedy, Drama
 39 Art House & International, Drama
 40 Comedy, Drama
 41 Comedy, Romance
 42 Drama, Kids & Family
 43 Action & Adventure, Classics, Drama, Western
 44 Drama, Romance
 45 Classics
 46 Classics, Comedy, Drama, Romance
 47 Drama, Mystery & Suspense
 48 Drama, Faith & Spirituality
 49 Drama, Mystery & Suspense
 50 Action & Adventure, Horror, Science Fiction & ...
 51 Horror, Mystery & Suspense
 52 Classics, Drama

53 Comedy
 54 Comedy, Drama
 55 Drama
 56 Horror, Mystery & Suspense
 57 Comedy, Science Fiction & Fantasy
 58 Drama
 59 Art House & International, Drama

directors \

0 Chris Columbus
 1 Nicole Holofcener
 2 Blake Edwards
 3 Sidney Lumet
 4 Richard Fleischer
 5 Roland Emmerich
 6 Alfred Hitchcock
 7 Delmer Daves
 8 Adam Thomas Anderegg
 9 D.W. Griffith
 10 Hideo Nakata
 11 Jonathan Kaplan
 12 Andy Garcia
 13 Michael Curtiz
 14 George Cukor
 15 Mary McGuckian
 16 Joseph Losey
 17 Gregory Jacobs, Gregory Jacobs, Greg Jacobs
 18 Will Vinton
 19 Andy Byatt, Alastair Fothergill
 20 Michael Curtiz, William Keighley
 21 Fritz Lang
 22 Jean-Baptiste Andrea, Fabrice Canepa, Jean-Bap...
 23 John Schultz
 24 MarÃa LidÃ³n
 25 Kevin Reynolds
 26 SÃbastien Lifshitz
 27 Vincent Sassone
 28 Wayne Kramer
 29 Mark Griffiths
 30 Kurt Wimmer
 31 Paola di Florio
 32 Luc Besson
 33 Stevan Mena
 34 Wilhelm Liebenberg, Federico Sanchez
 35 Charles Vidor, John Hudson, John Huston
 36 Richard Fleischer
 37 Bob Yari, Richard Shepard

38 Jeff Kramer, Jeffrey Kramer
39 Claire Denis
40 Michael McGowan
41 Michael Showalter
42 Roman Polanski
43 John Wayne
44 Christopher Scott Cherot
45 Robert Rossen
46 Lewis Gilbert
47 Joshua Michael Stern
48 Steve Taylor
49 Paula Van der Oest
50 James Cameron
51 Sebastian Gutierrez
52 Joseph L. Mankiewicz
53 Salvador Litvak
54 George Bamber
55 Chris Fisher
56 William Brent Bell
57 Carl Reiner
58 Delbert Mann
59 Jason Matzner

authors \

0 Craig Titley, Chris Columbus, Rick Riordan
1 Nicole Holofcener
2 Blake Edwards
3 Reginald Rose
4 Earl Felton
5 Harald Kloser, Roland Emmerich
6 Alma Reville, Charles Bennett, Ian Hay, Alfred...
7 Halsted Welles
8 Jack Weyland, Janine Whetton Gilbert
9 Gerrit J. Lloyd, Stephen Vincent Ben  t
10 Hideo Hakata, Takashige Ichise, Yoshihiro Naka...
11 Tom Topor
12 G. Cabrera Infante, Guillermo Cabrera Infante
13 Ranald MacDougall
14 Garson Kanin, Ruth Gordon
15 Mary McGuckian
16 Robert Thoeren, Hugo Butler, Dalton Trumbo
17 Gregory Jacobs, Fabi  n Bielinsky, Steven Sode...
18 Susan Shadburne
19 Alastair Fothergill, Andy Byatt, Tim Ecott
20 Norman Reilly Raine, Seton I. Miller, Michael ...
21 Dudley Nichols
22 Jean-Baptiste Andrea, Fabrice Canepa

23 John Schultz
 24 Adela Ibañez, Isabel Pisano
 25 Dean Georgaris
 26 Stéphanie Lifshitz, Stéphanie Bouquet
 27 Vincent Sassone
 28 Wayne Kramer
 29 Eric Goodman, Kelly Senecal
 30 Kurt Wimmer
 31 Paola di Florio
 32 Luc Besson, Celine Garcia
 33 Stevan Mena
 34 Wilhelm Liebenberg, Federico Sanchez
 35 Ben Hecht
 36 Earl Felton, Martin Goldsmith
 37 Richard Shepard, Richard Shepherd
 38 Jeff Kramer, Jeffrey Kramer
 39 Claire Denis, Jean-Pol Fargeau
 40 Michael McGowan
 41 Michael Showalter
 42 Ronald Harwood
 43 James Edward Grant
 44 Charles E. Drew, Christopher Scott Cherot, Cha...
 45 Robert Rossen
 46 Bill Naughton
 47 Joshua Michael Stern
 48 Steve Taylor, Chip Arnold, Ben Pearson, Henry ...
 49 Carel Donck
 50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak, Nina Davidovich
 54 David Vernon
 55 Chris Fisher, Gil Reavill, Eric Saks
 56 William Brent Bell, Matthew Peterman
 57 Henry Olek, Carl Reiner, Phil Alden Robinson
 58 Paul Monash
 59 Tom Willett, Jason Matzner

	actors	streaming_release_date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2015-11-25
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2012-09-04
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	2014-07-24
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	2017-01-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	2016-06-10
5	Steven Strait, Camilla Belle, Cliff Curtis, Jo...	2013-06-22
6	Robert Donat, Madeleine Carroll, Godfrey Tearl...	2017-01-12
7	Glenn Ford, Van Heflin, Felicia Farr, Leora Da...	2012-04-16

8	Heather Beers, Gary Neilson, Lisa McCammon, Ja...	2017-05-22
9	Walter Huston, Una Merkel, Kay Hammond, Ian Ke...	2013-12-03
10	Hitomi Kuroki, Rio Kanno, Shigemitsu Ogi, Mire...	2017-03-23
11	Jodie Foster, Kelly McGillis, Bernie Coulson, ...	2016-10-04
12	Andy Garcia, Dustin Hoffman, Bill Murray, InÃ©...	2017-03-23
13	John Garfield, Patricia Neal, Phyllis Thaxter,...	2016-08-25
14	Spencer Tracy, Katharine Hepburn, Judy Hollida...	2008-05-01
15	Gabriel Byrne, F. Murray Abraham, Kathy Bates,...	2014-07-24
16	Van Heflin, Evelyn Keyes, Katherine Warren, Jo...	2016-12-05
17	John C. Reilly, Diego Luna, Maggie Gyllenhaal,...	2016-04-27
18	James Whitmore, Michele Mariana, Gary Krug, Ch...	2017-02-04
19	Pierce Brosnan, Michael Gambon	2016-12-07
20	Errol Flynn, Olivia de Havilland, Basil Rathbo...	2014-01-23
21	Walter Pidgeon, Joan Bennett, George Sanders, ...	2014-03-01
22	Ray Wise, Lin Shaye, Mick Cain, Alexandra Hold...	2018-12-11
23	Matthew Hennessey, Kevin Corrigan, Steve Parla...	NaT
24	Daryl Hannah, Denise Richards, Joaquim de Alme...	2014-02-04
25	James Franco, Sophia Myles, Rufus Sewell, Davi...	2015-11-25
26	StÃ©phanie Michelini, Josiane Stoleru, Yasmine...	2005-10-25
27	Vincent Pastore, Vincent Sassone, Frank Vincen...	2018-01-16
28	Paul Walker, Cameron Bright, Vera Farmiga, Kar...	2008-10-01
29	Christopher Jacot, Shawn Roberts, Ryan Bellevi...	2017-03-10
30	Milla Jovovich, Cameron Bright, Nick Chinlund,...	2012-04-16
31	Mary Liuzzo, Tony Liuzzo, Sally Liuzzo, Penny ...	2016-04-03
32	Freddie Highmore, Mia Farrow, Penny Balfour, D...	2016-08-10
33	Samantha Dark, R. Brandon Johnson, Heather Mag...	2017-03-30
34	Sarah Manninen, Caroline Neron, Victoria Sanch...	2016-09-23
35	Rock Hudson, Jennifer Jones, Vittorio De Sica,...	2012-09-18
36	Charlie MacGraw, Charles McGraw, Marie Windsor...	2012-08-20
37	Pierce Brosnan, Greg Kinnear, Hope Davis, Phil...	2016-11-18
38	Mika Boorem, Wang Luoyong, Yi Ding (II), Beau ...	2016-11-18
39	Beatrice Dalle, Michael Subor, Jean-Marc Terii...	2006-04-25
40	Adam Butcher, Campbell Scott, Gordon Pinsent, ...	2017-05-11
41	Michael Showalter, Elizabeth Banks, Michelle W...	2018-07-02
42	Ben Kingsley, Barney Clark, Leanne Rowe, Mark ...	2012-04-16
43	John Wayne, Richard Widmark, Laurence Harvey, ...	2016-11-30
44	Richard T. Jones, Blair Underwood, Chenoa Maxw...	2018-04-10
45	Richard Burton, Fredric March, Claire Bloom, D...	2017-01-17
46	Michael Caine, Millicent Martin, Julia Foster,...	2013-08-01
47	Aaron Eckhart, Ian McKellen, Brittany Murphy, ...	2016-04-19
48	Michael W. Smith, Jeff Obafemi Carr, Jonathan ...	2006-07-18
49	Laurien Van Den Broeck, Hunter Bussemaker, Joh...	2020-03-13
50	Sigourney Weaver, Carrie Henn, Michael Biehn, ...	2015-11-25
51	Lucy Liu, Michael Chiklis, Carla Gugino, James...	2016-11-01
52	Bette Davis, Anne Baxter, Gary Merrill, George...	2015-11-25
53	Michael Lerner, Lesley Ann Warren, Jack Klugma...	2017-05-22
54	Daniel Letterle, Meredith Baxter, David Monaha...	2017-02-02

55	Cuba Gooding Jr., Wyclef Jean, Keith David, Cl...	2012-04-16
56	Jon Foster, Frankie Muniz, Samaire Armstrong, ...	2016-08-09
57	Steve Martin, Lily Tomlin, Victoria Tennant, M...	2016-10-30
58	Richard Thomas, Ernest Borgnine, Ian Holm, Don...	2018-04-16
59	Agnes Bruckner, Kelli Garner, Justin Long, Gin...	2012-04-16

	runtime	tomatometer_rating	tomatometer_count	audience_rating	\
0	119.0	49.0	149.0	53.0	
1	90.0	87.0	142.0	64.0	
2	122.0	67.0	24.0	53.0	
3	95.0	100.0	54.0	97.0	
4	127.0	89.0	27.0	74.0	
5	109.0	8.0	149.0	37.0	
6	80.0	96.0	51.0	86.0	
7	92.0	96.0	28.0	79.0	
8	103.0	20.0	10.0	87.0	
9	97.0	82.0	11.0	40.0	
10	100.0	80.0	15.0	66.0	
11	110.0	91.0	22.0	79.0	
12	143.0	25.0	83.0	64.0	
13	97.0	100.0	10.0	86.0	
14	101.0	96.0	28.0	86.0	
15	124.0	4.0	24.0	35.0	
16	92.0	100.0	18.0	86.0	
17	87.0	69.0	124.0	57.0	
18	86.0	80.0	5.0	82.0	
19	90.0	67.0	52.0	80.0	
20	102.0	100.0	48.0	89.0	
21	100.0	92.0	12.0	75.0	
22	83.0	75.0	8.0	60.0	
23	103.0	80.0	5.0	91.0	
24	77.0	0.0	6.0	63.0	
25	126.0	31.0	121.0	71.0	
26	94.0	63.0	16.0	80.0	
27	82.0	38.0	8.0	NaN	
28	121.0	41.0	131.0	79.0	
29	93.0	0.0	5.0	61.0	
30	88.0	8.0	83.0	30.0	
31	75.0	93.0	15.0	48.0	
32	94.0	22.0	92.0	56.0	
33	90.0	33.0	18.0	83.0	
34	108.0	21.0	34.0	39.0	
35	150.0	0.0	12.0	37.0	
36	71.0	100.0	11.0	88.0	
37	97.0	75.0	156.0	65.0	
38	107.0	15.0	13.0	66.0	
39	130.0	87.0	30.0	73.0	

40	98.0	64.0	69.0	85.0
41	91.0	32.0	75.0	75.0
42	130.0	61.0	143.0	63.0
43	192.0	52.0	21.0	63.0
44	96.0	24.0	29.0	60.0
45	141.0	0.0	8.0	42.0
46	114.0	96.0	27.0	76.0
47	102.0	14.0	7.0	65.0
48	104.0	44.0	16.0	78.0
49	90.0	60.0	10.0	67.0
50	137.0	97.0	76.0	94.0
51	97.0	36.0	14.0	20.0
52	138.0	99.0	77.0	94.0
53	87.0	42.0	38.0	68.0
54	88.0	30.0	27.0	58.0
55	97.0	21.0	28.0	42.0
56	85.0	10.0	59.0	55.0
57	93.0	85.0	41.0	67.0
58	156.0	100.0	5.0	69.0
59	88.0	67.0	15.0	65.0

	audience_count	tomatometer_top_critics_count \
0	254421.0	43
1	11574.0	44
2	14684.0	2
3	105386.0	6
4	68918.0	5
5	411140.0	37
6	23890.0	8
7	9243.0	6
8	4819.0	0
9	457.0	4
10	21475.0	3
11	20821.0	1
12	25944.0	37
13	335.0	1
14	10563.0	6
15	1935.0	13
16	286.0	6
17	6711.0	35
18	1046.0	1
19	4692.0	18
20	33946.0	8
21	554.0	0
22	10373.0	1
23	386.0	1
24	914.0	4

25	220956.0	36
26	867.0	4
27	NaN	5
28	135400.0	39
29	13033.0	2
30	279994.0	21
31	866.0	7
32	134483.0	30
33	2622.0	4
34	1295.0	10
35	1460.0	6
36	1135.0	0
37	73701.0	39
38	2059.0	6
39	9701.0	10
40	6203.0	26
41	7436.0	25
42	23294.0	39
43	7337.0	4
44	1143.0	17
45	5383.0	1
46	14669.0	6
47	4059.0	1
48	79542.0	8
49	1430.0	6
50	429674.0	17
51	8239.0	2
52	45248.0	22
53	1121.0	19
54	2394.0	13
55	7326.0	10
56	113424.0	18
57	14346.0	10
58	7588.0	0
59	5650.0	5

	tomatometer_fresh_critics_count	tomatometer_rotten_critics_count
0	73	76
1	123	19
2	16	8
3	54	0
4	24	3
5	12	137
6	49	2
7	27	1
8	2	8
9	9	2

10	12	3
11	20	2
12	21	62
13	10	0
14	27	1
15	1	23
16	18	0
17	85	39
18	4	1
19	35	17
20	48	0
21	11	1
22	6	2
23	4	1
24	0	6
25	38	83
26	10	6
27	3	5
28	54	77
29	0	5
30	7	76
31	14	1
32	20	72
33	6	12
34	7	27
35	0	12
36	11	0
37	117	39
38	2	11
39	26	4
40	44	25
41	24	51
42	87	56
43	11	10
44	7	22
45	0	8
46	26	1
47	1	6
48	7	9
49	6	4
50	74	2
51	5	9
52	76	1
53	16	22
54	8	19
55	6	22
56	6	53

57	35	6
58	5	0
59	10	5

4.2 2. Perform any data extraction/selection steps.

```
[33]: # Count the number of missing (NaN) values in the cleaned DataFrame
nan_summary = movies_df_cleaned.isnull().sum()
print(nan_summary)
```

```
movie_title           0
critics_consensus    8578
content_rating        0
genres                19
directors             194
authors              1542
actors                352
streaming_release_date 384
runtime              314
tomatometer_rating    44
tomatometer_count     44
audience_rating      296
audience_count       297
tomatometer_top_critics_count 0
tomatometer_fresh_critics_count 0
tomatometer_rotten_critics_count 0
dtype: int64
```

```
[34]: movies_df_cleaned.dtypes
```

```
[34]: movie_title           object
critics_consensus         object
content_rating            object
genres                   object
directors                 object
authors                  object
actors                   object
streaming_release_date    datetime64[ns]
runtime                   float64
tomatometer_rating        float64
tomatometer_count         float64
audience_rating          float64
audience_count           float64
tomatometer_top_critics_count  int64
tomatometer_fresh_critics_count  int64
tomatometer_rotten_critics_count  int64
dtype: object
```

4.3 3. Transform features if necessary.

```
[ ]:
```

4.4 4. Engineer new useful features.

```
[ ]:
```

4.5 5. Deal with missing data (do not just drop rows or columns without justifying this).

```
[35]: # Replacing NaN values in the specified 'object' columns with 'Unknown'
columns_to_fill = ['movie_title', 'critics_consensus', 'content_rating',
                  ↪ 'genres', 'directors', 'authors', 'actors']
movies_df_cleaned[columns_to_fill] = movies_df_cleaned[columns_to_fill].
                  ↪ fillna('Unknown')
```

```
[36]: # Replacing NaN values in the specified 'float64' columns with 0
columns_to_fill_numeric = ['runtime', 'tomatometer_rating',
                          ↪ 'tomatometer_count', 'audience_rating', 'audience_count']
movies_df_cleaned[columns_to_fill_numeric] =
                  ↪ movies_df_cleaned[columns_to_fill_numeric].fillna(0)
```

```
[37]: # Fill NaN values in 'streaming_release_date' with a default date of January 1,
                  ↪ 1800.
movies_df_cleaned['streaming_release_date'] =
                  ↪ movies_df_cleaned['streaming_release_date'].fillna(pd.
                  ↪ Timestamp('1800-01-01'))
```

4.6 6. Create dummy variables if necessary.

```
[38]: # Creating dummy variables for the 'content_rating' column in the dataset
                  ↪ 'movies_df_cleaned'.
movies_df_dummies = pd.get_dummies(movies_df_cleaned['content_rating'],
                  ↪ prefix='rating_type', dtype = int)

# Concatenate the dummy variables to the original DataFrame.
movies_df_cleaned = pd.concat([movies_df_cleaned, movies_df_dummies], axis=1)
```

```
[39]: movies_df_cleaned.shape
```

```
[39]: (17712, 22)
```

```
[40]: movies_df_cleaned.dtypes
```

```
[40]: movie_title          object
      critics_consensus  object
```

```

content_rating          object
genres                  object
directors               object
authors                 object
actors                  object
streaming_release_date  datetime64[ns]
runtime                 float64
tomatometer_rating      float64
tomatometer_count       float64
audience_rating        float64
audience_count         float64
tomatometer_top_critics_count  int64
tomatometer_fresh_critics_count int64
tomatometer_rotten_critics_count int64
rating_type_G           int32
rating_type_NC17        int32
rating_type_NR          int32
rating_type_PG          int32
rating_type_PG-13       int32
rating_type_R           int32
dtype: object

```

```
[41]: movies_df_cleaned.head(60)
```

```

[41]:
      movie_title \
0  Percy Jackson & the Olympians: The Lightning T...
1                Please Give
2                                10
3      12 Angry Men (Twelve Angry Men)
4      20,000 Leagues Under The Sea
5                10,000 B.C.
6                The 39 Steps
7                3:10 to Yuma
8      Charly (A Heartbeat Away)
9      Abraham Lincoln
10             Dark Water
11             The Accused
12             The Lost City
13             The Breaking Point
14             Adam's Rib
15             The Bridge of San Luis Rey
16             The Prowler (Cost of Living )
17             Criminal
18             The Adventures of Mark Twain
19             Deep Blue
20             The Adventures of Robin Hood
21             Man Hunt

```

22	Dead End
23	Bandwagon
24	Whore
25	Tristan & Isolde
26	Wild Side
27	A Tale of Two Pizzas
28	Running Scared
29	Going the Distance
30	Ultraviolet
31	Home of the Brave
32	Arthur and the Invisibles (Arthur and the Mini...
33	Malevolence
34	Eternal
35	A Farewell To Arms
36	The Narrow Margin
37	The Matador
38	Smile
39	The Intruder (L'Intrus)
40	Saint Ralph
41	The Baxter
42	Oliver Twist
43	The Alamo
44	G
45	Alexander the Great
46	Alfie
47	Neverwas
48	The Second Chance
49	Moonlight
50	Aliens
51	Rise: Blood Hunter
52	All About Eve
53	When Do We Eat?
54	The Mostly Unfabulous Social Life of Ethan Green
55	Dirty
56	Stay Alive
57	All of Me
58	All Quiet on the Western Front
59	Dreamland

	critics_consensus	content_rating	\
0	Though it may seem like just another Harry Pot...	PG	
1	Nicole Holofcener's newest might seem slight i...	R	
2	Blake Edwards' bawdy comedy may not score a pe...	R	
3	Sidney Lumet's feature debut is a superbly wri...	NR	
4	One of Disney's finest live-action adventures,...	G	
5	With attention strictly paid to style instead ...	PG-13	
6	Packed with twists and turns, this essential e...	NR	

7	Unknown	NR
8	Unknown	PG
9	Unknown	NR
10	Unknown	PG-13
11	Unknown	R
12	Its heart is in the right place, but what star...	R
13	Unknown	NR
14	Matched by Garson Kanin's witty, sophisticated...	NR
15	Despite an all-star cast and some impressive v...	PG
16	Unknown	PG
17	If you saw Nine Queens, it may feel redundant,...	R
18	Unknown	G
19	Full of visually striking footage, Deep Blue m...	G
20	Errol Flynn thrills as the legendary title cha...	PG
21	Unknown	NR
22	Unknown	R
23	Unknown	R
24	Unknown	R
25	Competent but somewhat static, Tristan & Isold...	PG-13
26	Unknown	NR
27	Unknown	PG
28	This film runs with frenetic energy punctuated...	R
29	Unknown	R
30	An incomprehensible and forgettable sci-fi thr...	PG-13
31	Unknown	R
32	Arthur wastes its big-name voice talent on a p...	PG
33	Unknown	R
34	Even by the standards of camp, Eternal fails t...	R
35	Unknown	NR
36	Unknown	NR
37	This humorously amoral, oddball comic thriller...	R
38	Unknown	PG-13
39	The impressionistic narrative may confound the...	G
40	Saint Ralph is an amiable tear-jerker that wal...	PG-13
41	The Baxter is good-natured, but there are simp...	PG-13
42	Polanski's version of Dickens' classic won't h...	PG-13
43	John Wayne proves to be less compelling behind...	NR
44	Race and class struggles come to the Hamptons ...	R
45	Unknown	G
46	Anchored by Michael Caine's charmingly droll p...	PG
47	Unknown	PG-13
48	Unknown	PG-13
49	Unknown	R
50	While Alien was a marvel of slow-building, atm...	R
51	Unknown	R
52	Smart, sophisticated, and devastatingly funny,...	PG
53	A crass, sitcom-ish holiday comedy, When Do We...	R

54	Ethan Green makes for an annoying protagonist,...	R
55	Dirty is an unoriginal L.A. cop drama that wea...	R
56	A by-the-numbers teen horror flick, Stay Alive...	PG-13
57	A high-concept farce carried by Carl Reiner's ...	PG
58	Unknown	NR
59	Unknown	PG-13

		genres \
0	Action & Adventure, Comedy, Drama, Science Fic...	
1	Comedy	
2	Comedy, Romance	
3	Classics, Drama	
4	Action & Adventure, Drama, Kids & Family	
5	Action & Adventure, Classics, Drama	
6	Action & Adventure, Classics, Mystery & Suspense	
7	Classics, Drama, Western	
8	Comedy, Drama, Romance	
9	Classics, Drama	
10	Art House & International, Horror, Mystery & S...	
11	Drama, Mystery & Suspense	
12	Drama	
13	Drama	
14	Classics, Comedy, Romance	
15	Art House & International, Drama, Faith & Spir...	
16	Drama, Mystery & Suspense	
17	Drama, Mystery & Suspense	
18	Action & Adventure, Animation, Kids & Family	
19	Documentary, Special Interest	
20	Action & Adventure, Classics, Romance	
21	Classics, Drama, Mystery & Suspense	
22	Comedy, Horror, Mystery & Suspense	
23	Comedy	
24	Art House & International, Drama	
25	Drama, Science Fiction & Fantasy, Romance	
26	Art House & International, Drama	
27	Comedy	
28	Action & Adventure, Drama, Mystery & Suspense	
29	Comedy	
30	Action & Adventure, Drama, Horror, Science Fic...	
31	Action & Adventure, Documentary, Drama, Specia...	
32	Animation, Kids & Family, Science Fiction & Fa...	
33	Horror	
34	Horror, Mystery & Suspense	
35	Classics, Drama, Romance	
36	Classics, Drama, Mystery & Suspense	
37	Action & Adventure, Comedy, Drama, Mystery & S...	
38	Art House & International, Comedy, Drama	

39 Art House & International, Drama
 40 Comedy, Drama
 41 Comedy, Romance
 42 Drama, Kids & Family
 43 Action & Adventure, Classics, Drama, Western
 44 Drama, Romance
 45 Classics
 46 Classics, Comedy, Drama, Romance
 47 Drama, Mystery & Suspense
 48 Drama, Faith & Spirituality
 49 Drama, Mystery & Suspense
 50 Action & Adventure, Horror, Science Fiction & ...
 51 Horror, Mystery & Suspense
 52 Classics, Drama
 53 Comedy
 54 Comedy, Drama
 55 Drama
 56 Horror, Mystery & Suspense
 57 Comedy, Science Fiction & Fantasy
 58 Drama
 59 Art House & International, Drama

directors \
 0 Chris Columbus
 1 Nicole Holofcener
 2 Blake Edwards
 3 Sidney Lumet
 4 Richard Fleischer
 5 Roland Emmerich
 6 Alfred Hitchcock
 7 Delmer Daves
 8 Adam Thomas Anderegg
 9 D.W. Griffith
 10 Hideo Nakata
 11 Jonathan Kaplan
 12 Andy Garcia
 13 Michael Curtiz
 14 George Cukor
 15 Mary McGuckian
 16 Joseph Losey
 17 Gregory Jacobs, Gregory Jacobs, Greg Jacobs
 18 Will Vinton
 19 Andy Byatt, Alastair Fothergill
 20 Michael Curtiz, William Keighley
 21 Fritz Lang
 22 Jean-Baptiste Andrea, Fabrice Canepa, Jean-Bap...
 23 John Schultz

24 MarÃa LidÃ³n
 25 Kevin Reynolds
 26 SÃbastien Lifshitz
 27 Vincent Sassone
 28 Wayne Kramer
 29 Mark Griffiths
 30 Kurt Wimmer
 31 Paola di Florio
 32 Luc Besson
 33 Stevan Mena
 34 Wilhelm Liebenberg, Federico Sanchez
 35 Charles Vidor, John Hudson, John Huston
 36 Richard Fleischer
 37 Bob Yari, Richard Shepard
 38 Jeff Kramer, Jeffrey Kramer
 39 Claire Denis
 40 Michael McGowan
 41 Michael Showalter
 42 Roman Polanski
 43 John Wayne
 44 Christopher Scott Cherot
 45 Robert Rossen
 46 Lewis Gilbert
 47 Joshua Michael Stern
 48 Steve Taylor
 49 Paula Van der Oest
 50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak
 54 George Bamber
 55 Chris Fisher
 56 William Brent Bell
 57 Carl Reiner
 58 Delbert Mann
 59 Jason Matzner

authors \
 0 Craig Titley, Chris Columbus, Rick Riordan
 1 Nicole Holofcener
 2 Blake Edwards
 3 Reginald Rose
 4 Earl Felton
 5 Harald Kloser, Roland Emmerich
 6 Alma Reville, Charles Bennett, Ian Hay, Alfred..
 7 Halsted Welles
 8 Jack Weyland, Janine Whetton Gilbert

9 Gerrit J. Lloyd, Stephen Vincent Ben t
 10 Hideo Hakata, Takashige Ichise, Yoshihiro Naka...
 11 Tom Topor
 12 G. Cabrera Infante, Guillermo Cabrera Infante
 13 Ranald MacDougall
 14 Garson Kanin, Ruth Gordon
 15 Mary McGuckian
 16 Robert Thoeren, Hugo Butler, Dalton Trumbo
 17 Gregory Jacobs, Fabi n Bielinsky, Steven Sode...
 18 Susan Shadburne
 19 Alastair Fothergill, Andy Byatt, Tim Ecott
 20 Norman Reilly Raine, Seton I. Miller, Michael ...
 21 Dudley Nichols
 22 Jean-Baptiste Andrea, Fabrice Canepa
 23 John Schultz
 24 Adela Iba ez, Isabel Pisano
 25 Dean Georgaris
 26 S bastien Lifshitz, St phane Bouquet
 27 Vincent Sassone
 28 Wayne Kramer
 29 Eric Goodman, Kelly Senecal
 30 Kurt Wimmer
 31 Paola di Florio
 32 Luc Besson, Celine Garcia
 33 Stevan Mena
 34 Wilhelm Liebenberg, Federico Sanchez
 35 Ben Hecht
 36 Earl Felton, Martin Goldsmith
 37 Richard Shepard, Richard Shepherd
 38 Jeff Kramer, Jeffrey Kramer
 39 Claire Denis, Jean-Pol Fargeau
 40 Michael McGowan
 41 Michael Showalter
 42 Ronald Harwood
 43 James Edward Grant
 44 Charles E. Drew, Christopher Scott Cherot, Cha...
 45 Robert Rossen
 46 Bill Naughton
 47 Joshua Michael Stern
 48 Steve Taylor, Chip Arnold, Ben Pearson, Henry ...
 49 Carel Donck
 50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak, Nina Davidovich
 54 David Vernon
 55 Chris Fisher, Gil Reavill, Eric Saks

56 William Brent Bell, Matthew Peterman
 57 Henry Olek, Carl Reiner, Phil Alden Robinson
 58 Paul Monash
 59 Tom Willett, Jason Matzner

	actors	streaming_release_date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2015-11-25
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2012-09-04
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	2014-07-24
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	2017-01-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	2016-06-10
5	Steven Strait, Camilla Belle, Cliff Curtis, Jo...	2013-06-22
6	Robert Donat, Madeleine Carroll, Godfrey Tearl...	2017-01-12
7	Glenn Ford, Van Heflin, Felicia Farr, Leora Da...	2012-04-16
8	Heather Beers, Gary Neilson, Lisa McCammon, Ja...	2017-05-22
9	Walter Huston, Una Merkel, Kay Hammond, Ian Ke...	2013-12-03
10	Hitomi Kuroki, Rio Kanno, Shigemitsu Ogi, Mire...	2017-03-23
11	Jodie Foster, Kelly McGillis, Bernie Coulson, ...	2016-10-04
12	Andy Garcia, Dustin Hoffman, Bill Murray, InÃ©...	2017-03-23
13	John Garfield, Patricia Neal, Phyllis Thaxter,...	2016-08-25
14	Spencer Tracy, Katharine Hepburn, Judy Hollida...	2008-05-01
15	Gabriel Byrne, F. Murray Abraham, Kathy Bates,...	2014-07-24
16	Van Heflin, Evelyn Keyes, Katherine Warren, Jo...	2016-12-05
17	John C. Reilly, Diego Luna, Maggie Gyllenhaal,...	2016-04-27
18	James Whitmore, Michele Mariana, Gary Krug, Ch...	2017-02-04
19	Pierce Brosnan, Michael Gambon	2016-12-07
20	Errol Flynn, Olivia de Havilland, Basil Rathbo...	2014-01-23
21	Walter Pidgeon, Joan Bennett, George Sanders, ...	2014-03-01
22	Ray Wise, Lin Shaye, Mick Cain, Alexandra Hold...	2018-12-11
23	Matthew Hennessey, Kevin Corrigan, Steve Parla...	1800-01-01
24	Daryl Hannah, Denise Richards, Joaquim de Alme...	2014-02-04
25	James Franco, Sophia Myles, Rufus Sewell, Davi...	2015-11-25
26	StÃ©phanie Michelini, Josiane Stoleru, Yasmine...	2005-10-25
27	Vincent Pastore, Vincent Sassone, Frank Vincen...	2018-01-16
28	Paul Walker, Cameron Bright, Vera Farmiga, Kar...	2008-10-01
29	Christopher Jacot, Shawn Roberts, Ryan Bellevi...	2017-03-10
30	Milla Jovovich, Cameron Bright, Nick Chinlund,...	2012-04-16
31	Mary Liuzzo, Tony Liuzzo, Sally Liuzzo, Penny ...	2016-04-03
32	Freddie Highmore, Mia Farrow, Penny Balfour, D...	2016-08-10
33	Samantha Dark, R. Brandon Johnson, Heather Mag...	2017-03-30
34	Sarah Manninen, Caroline Neron, Victoria Sanch...	2016-09-23
35	Rock Hudson, Jennifer Jones, Vittorio De Sica,...	2012-09-18
36	Charlie MacGraw, Charles McGraw, Marie Windsor...	2012-08-20
37	Pierce Brosnan, Greg Kinnear, Hope Davis, Phil...	2016-11-18
38	Mika Boorem, Wang Luoyong, Yi Ding (II), Beau ...	2016-11-18
39	Beatrice Dalle, Michael Subor, Jean-Marc Terii...	2006-04-25
40	Adam Butcher, Campbell Scott, Gordon Pinsent, ...	2017-05-11

41	Michael Showalter, Elizabeth Banks, Michelle W...	2018-07-02
42	Ben Kingsley, Barney Clark, Leanne Rowe, Mark ...	2012-04-16
43	John Wayne, Richard Widmark, Laurence Harvey, ...	2016-11-30
44	Richard T. Jones, Blair Underwood, Chenoa Maxw...	2018-04-10
45	Richard Burton, Fredric March, Claire Bloom, D...	2017-01-17
46	Michael Caine, Millicent Martin, Julia Foster,...	2013-08-01
47	Aaron Eckhart, Ian McKellen, Brittany Murphy, ...	2016-04-19
48	Michael W. Smith, Jeff Obafemi Carr, Jonathan ...	2006-07-18
49	Laurien Van Den Broeck, Hunter Bussemaker, Joh...	2020-03-13
50	Sigourney Weaver, Carrie Henn, Michael Biehn, ...	2015-11-25
51	Lucy Liu, Michael Chiklis, Carla Gugino, James...	2016-11-01
52	Bette Davis, Anne Baxter, Gary Merrill, George...	2015-11-25
53	Michael Lerner, Lesley Ann Warren, Jack Klugma...	2017-05-22
54	Daniel Letterle, Meredith Baxter, David Monaha...	2017-02-02
55	Cuba Gooding Jr., Wyclef Jean, Keith David, Cl...	2012-04-16
56	Jon Foster, Frankie Muniz, Samaire Armstrong, ...	2016-08-09
57	Steve Martin, Lily Tomlin, Victoria Tennant, M...	2016-10-30
58	Richard Thomas, Ernest Borgnine, Ian Holm, Don...	2018-04-16
59	Agnes Bruckner, Kelli Garner, Justin Long, Gin...	2012-04-16

	runtime	tomatometer_rating	...	audience_count	\
0	119.0	49.0	...	254421.0	
1	90.0	87.0	...	11574.0	
2	122.0	67.0	...	14684.0	
3	95.0	100.0	...	105386.0	
4	127.0	89.0	...	68918.0	
5	109.0	8.0	...	411140.0	
6	80.0	96.0	...	23890.0	
7	92.0	96.0	...	9243.0	
8	103.0	20.0	...	4819.0	
9	97.0	82.0	...	457.0	
10	100.0	80.0	...	21475.0	
11	110.0	91.0	...	20821.0	
12	143.0	25.0	...	25944.0	
13	97.0	100.0	...	335.0	
14	101.0	96.0	...	10563.0	
15	124.0	4.0	...	1935.0	
16	92.0	100.0	...	286.0	
17	87.0	69.0	...	6711.0	
18	86.0	80.0	...	1046.0	
19	90.0	67.0	...	4692.0	
20	102.0	100.0	...	33946.0	
21	100.0	92.0	...	554.0	
22	83.0	75.0	...	10373.0	
23	103.0	80.0	...	386.0	
24	77.0	0.0	...	914.0	
25	126.0	31.0	...	220956.0	

26	94.0	63.0	...	867.0
27	82.0	38.0	...	0.0
28	121.0	41.0	...	135400.0
29	93.0	0.0	...	13033.0
30	88.0	8.0	...	279994.0
31	75.0	93.0	...	866.0
32	94.0	22.0	...	134483.0
33	90.0	33.0	...	2622.0
34	108.0	21.0	...	1295.0
35	150.0	0.0	...	1460.0
36	71.0	100.0	...	1135.0
37	97.0	75.0	...	73701.0
38	107.0	15.0	...	2059.0
39	130.0	87.0	...	9701.0
40	98.0	64.0	...	6203.0
41	91.0	32.0	...	7436.0
42	130.0	61.0	...	23294.0
43	192.0	52.0	...	7337.0
44	96.0	24.0	...	1143.0
45	141.0	0.0	...	5383.0
46	114.0	96.0	...	14669.0
47	102.0	14.0	...	4059.0
48	104.0	44.0	...	79542.0
49	90.0	60.0	...	1430.0
50	137.0	97.0	...	429674.0
51	97.0	36.0	...	8239.0
52	138.0	99.0	...	45248.0
53	87.0	42.0	...	1121.0
54	88.0	30.0	...	2394.0
55	97.0	21.0	...	7326.0
56	85.0	10.0	...	113424.0
57	93.0	85.0	...	14346.0
58	156.0	100.0	...	7588.0
59	88.0	67.0	...	5650.0

	tomatometer_top_critics_count	tomatometer_fresh_critics_count \
0	43	73
1	44	123
2	2	16
3	6	54
4	5	24
5	37	12
6	8	49
7	6	27
8	0	2
9	4	9
10	3	12

11	1	20
12	37	21
13	1	10
14	6	27
15	13	1
16	6	18
17	35	85
18	1	4
19	18	35
20	8	48
21	0	11
22	1	6
23	1	4
24	4	0
25	36	38
26	4	10
27	5	3
28	39	54
29	2	0
30	21	7
31	7	14
32	30	20
33	4	6
34	10	7
35	6	0
36	0	11
37	39	117
38	6	2
39	10	26
40	26	44
41	25	24
42	39	87
43	4	11
44	17	7
45	1	0
46	6	26
47	1	1
48	8	7
49	6	6
50	17	74
51	2	5
52	22	76
53	19	16
54	13	8
55	10	6
56	18	6
57	10	35

58
59

0
5

5
10

	tomatometer_rotten_critics_count	rating_type_G	rating_type_NC17	\
0	76	0	0	
1	19	0	0	
2	8	0	0	
3	0	0	0	
4	3	1	0	
5	137	0	0	
6	2	0	0	
7	1	0	0	
8	8	0	0	
9	2	0	0	
10	3	0	0	
11	2	0	0	
12	62	0	0	
13	0	0	0	
14	1	0	0	
15	23	0	0	
16	0	0	0	
17	39	0	0	
18	1	1	0	
19	17	1	0	
20	0	0	0	
21	1	0	0	
22	2	0	0	
23	1	0	0	
24	6	0	0	
25	83	0	0	
26	6	0	0	
27	5	0	0	
28	77	0	0	
29	5	0	0	
30	76	0	0	
31	1	0	0	
32	72	0	0	
33	12	0	0	
34	27	0	0	
35	12	0	0	
36	0	0	0	
37	39	0	0	
38	11	0	0	
39	4	1	0	
40	25	0	0	
41	51	0	0	
42	56	0	0	

43	10	0	0
44	22	0	0
45	8	1	0
46	1	0	0
47	6	0	0
48	9	0	0
49	4	0	0
50	2	0	0
51	9	0	0
52	1	0	0
53	22	0	0
54	19	0	0
55	22	0	0
56	53	0	0
57	6	0	0
58	0	0	0
59	5	0	0

	rating_type_NR	rating_type_PG	rating_type_PG-13	rating_type_R
0	0	1	0	0
1	0	0	0	1
2	0	0	0	1
3	1	0	0	0
4	0	0	0	0
5	0	0	1	0
6	1	0	0	0
7	1	0	0	0
8	0	1	0	0
9	1	0	0	0
10	0	0	1	0
11	0	0	0	1
12	0	0	0	1
13	1	0	0	0
14	1	0	0	0
15	0	1	0	0
16	0	1	0	0
17	0	0	0	1
18	0	0	0	0
19	0	0	0	0
20	0	1	0	0
21	1	0	0	0
22	0	0	0	1
23	0	0	0	1
24	0	0	0	1
25	0	0	1	0
26	1	0	0	0
27	0	1	0	0

28	0	0	0	1
29	0	0	0	1
30	0	0	1	0
31	0	0	0	1
32	0	1	0	0
33	0	0	0	1
34	0	0	0	1
35	1	0	0	0
36	1	0	0	0
37	0	0	0	1
38	0	0	1	0
39	0	0	0	0
40	0	0	1	0
41	0	0	1	0
42	0	0	1	0
43	1	0	0	0
44	0	0	0	1
45	0	0	0	0
46	0	1	0	0
47	0	0	1	0
48	0	0	1	0
49	0	0	0	1
50	0	0	0	1
51	0	0	0	1
52	0	1	0	0
53	0	0	0	1
54	0	0	0	1
55	0	0	0	1
56	0	0	1	0
57	0	1	0	0
58	1	0	0	0
59	0	0	1	0

[60 rows x 22 columns]

```
[42]: # Count the number of missing (NaN) values in the cleaned DataFrame
nan_summary = movies_df_cleaned.isnull().sum()
print(nan_summary)
```

movie_title	0
critics_consensus	0
content_rating	0
genres	0
directors	0
authors	0
actors	0
streaming_release_date	0
runtime	0

```

tomatometer_rating      0
tomatometer_count       0
audience_rating        0
audience_count         0
tomatometer_top_critics_count  0
tomatometer_fresh_critics_count  0
tomatometer_rotten_critics_count  0
rating_type_G          0
rating_type_NC17       0
rating_type_NR         0
rating_type_PG         0
rating_type_PG-13      0
rating_type_R          0
dtype: int64

```

5 Term Project Milestone 2 Process.

Data Loading and Initial Exploration: - The initial step involves loading the movie dataset and inspecting the first 60 rows. This step helps to get a quick overview of the data structure, column names, and the general content of the dataset.

Handling Missing Data: - Performed missing data checks using the `isnull().sum()` function to count how many missing (NaN) values exist in each column. This step is crucial in deciding how to handle missing data before proceeding with analysis. - The output shows that columns such as `movie_title`, `critics_consensus`, `content_rating`, `genres`, etc., have no missing values after data cleaning, but this check helps verify completeness.

Data Transformation (Replacing NaN): - Apply transformations to handle missing values in specific columns: - For text-based columns such as `movie_title`, `critics_consensus`, `content_rating`, etc., missing values are replaced with “Unknown” placeholder. - For numeric columns such as `runtime`, `tomatometer_rating`, `audience_rating`, missing values are typically replaced with 0 to represent a lack of data or a neutral/default state.

Creation of Dummy Variables: - Dummy variables are created for the `content_rating`, `tomatometer_status`, and `audience_status` columns. This step involves transforming categorical data into a one-hot encoding format where each unique category becomes a separate binary (0 or 1) column.

Final Data Inspection: - After the transformations and the creation of dummy variables, an inspection of the modified dataset again using `head()` to ensure that the transformations have been applied correctly. - Also perform a final check to ensure there are no remaining NaN values in the critical columns by running `isnull().sum()` again, verifying that all missing values have been handled.

Each step in the process is aimed at ensuring the dataset is clean, complete, and ready for analysis or modeling tasks, such as predicting audience scores or performing regression analysis.

6 Milestone 3: Model Building and Evaluation.

In Milestone 3, you will begin the process of selecting, building, and evaluating a model. You are required to train and evaluate at least one model in this milestone. Write step-by-step for performing each of these steps. You can use any methods/tools you think are most appropriate, but you should explain/justify why you are selecting the model(s) and evaluation metric(s) you choose. It is important to think about what type of model and metric makes sense for your problem. Again, do what makes the most sense for your project. Write a short overview/conclusion of the insights gained from your model building/evaluation.

It is important to note that these milestones are meant to keep you on track for the final project submission. At any point, you can pivot or modify your project as needed based on what you discover. These milestones are not final versions; they are drafts of the many steps you need to complete along the way.

```
[43]: # Import necessary libraries.
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

```
[44]: # Load the data.
data = pd.read_csv('cleaned_movie_data.csv')

# Preview the data.
data.head()
```

```
[44]:
```

		title	year	rating	release_date	\
0		The Shawshank Redemption	1994	R	10/14/1994	
1		The Godfather	1972	R	3/24/1972	
2		The Dark Knight	2008	PG-13	7/18/2008	
3		The Godfather Part II	1974	R	12/18/1974	
4		The Lord of the Rings: The Return of the King	2003	PG-13	12/17/2003	

	runtime	genre	director	\
0	142	Drama	Frank Darabont	
1	175	Crime, Drama	Francis Ford Coppola	
2	152	Action, Crime, Drama	Christopher Nolan	
3	202	Crime, Drama	Francis Ford Coppola	
4	201	Action, Adventure, Drama	Peter Jackson	

	writer	\
0	Stephen King, Frank Darabont	

```

1          Mario Puzo, Francis Ford Coppola
2 Jonathan Nolan, Christopher Nolan, David S. Goyer
3          Francis Ford Coppola, Mario Puzo
4          J.R.R. Tolkien, Fran Walsh, Philippa Boyens

```

```

                                actors \
0      Tim Robbins, Morgan Freeman, Bob Gunton
1      Marlon Brando, Al Pacino, James Caan
2 Christian Bale, Heath Ledger, Aaron Eckhart
3      Al Pacino, Robert De Niro, Robert Duvall
4      Elijah Wood, Viggo Mortensen, Ian McKellen

```

```

                                plot \
0 A banker convicted of uxoricide forms a friend...
1 The aging patriarch of an organized crime dyna...
2 When a menace known as the Joker wreaks havoc ...
3 The early life and career of Vito Corleone in ...
4 Gandalf and Aragorn lead the World of Men agai...

```

```

                                Country \
0          United States
1          United States
2 United States, United Kingdom
3          United States
4      New Zealand, United States

```

```

                                Awards  Metascore  imdbRating \
0 Nominated for 7 Oscars. 21 wins & 42 nominatio...      82          9.3
1      Won 3 Oscars. 31 wins & 31 nominations total      100          9.2
2      Won 2 Oscars. 164 wins & 164 nominations total      84          9.0
3      Won 6 Oscars. 17 wins & 21 nominations total       90          9.0
4      Won 11 Oscars. 215 wins & 124 nominations total      94          9.0

```

```

imdbVotes  imdbID  BoxOffice  rt_rating
0  2,957,050  tt0111161  $28,767,189      89%
1  2,057,473  tt0068646  $136,381,073     97%
2  2,932,729  tt0468569  $534,987,076     94%
3  1,392,629  tt0071562  $47,834,595     96%
4  2,024,305  tt0167260  $381,878,219     94%

```

```
[45]: data.dtypes
```

```

[45]: title          object
      year           int64
      rating         object
      release_date   object
      runtime        int64

```

```

genre            object
director         object
writer           object
actors           object
plot             object
Country          object
Awards           object
Metascore        int64
imdbRating        float64
imdbVotes         object
imdbID           object
BoxOffice         object
rt_rating         object
dtype: object

```

```

[46]: # Convert 'imdbVotes' from object to integer.
data['imdbVotes'] = data['imdbVotes'].astype(str)
# data['imdbVotes'] = data['imdbVotes'].str.replace(',', '').astype(int)

# Convert 'BoxOffice' from object to float.
data['BoxOffice'] = data['BoxOffice'].str.replace('$', '').str.replace(',', '').
    ↪astype(float)

# Convert 'rt_rating' from object to integer.
data['rt_rating'] = pd.to_numeric(data['rt_rating'].str.replace('%', ''),
    ↪errors='coerce').fillna(0).astype(int)

# Check the data types after conversion.
print(data.dtypes)

```

```

title            object
year             int64
rating           object
release_date     object
runtime          int64
genre            object
director         object
writer           object
actors           object
plot             object
Country          object
Awards           object
Metascore        int64
imdbRating        float64
imdbVotes         object
imdbID           object
BoxOffice         float64
rt_rating         int32

```

dtype: object

```
[47]: # Select relevant features.
features = ['genre', 'runtime', 'director', 'actors', 'rt_rating']
target = 'imdbRating'

# Drop rows with missing values in selected columns.
data = data[features + [target]].dropna()
```

```
[48]: print(data.columns)
```

```
Index(['genre', 'runtime', 'director', 'actors', 'rt_rating', 'imdbRating'],
      dtype='object')
```

```
[49]: data.dtypes
```

```
[49]: genre          object
runtime         int64
director        object
actors          object
rt_rating       int32
imdbRating      float64
dtype: object
```

```
[50]: # Define feature categories for preprocessing.
numeric_features = ['runtime', 'rt_rating']
categorical_features = ['genre', 'director', 'actors']
```

```
[51]: # Define the preprocessing pipeline.
numeric_transformer = 'passthrough' # Numeric features don't need scaling for
↳ now
categorical_transformer = OneHotEncoder(drop='first', handle_unknown='ignore')

# Create a ColumnTransformer for applying transformations.
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, numeric_features), # Numeric processing.
        ('cat', categorical_transformer, categorical_features) # Categorical
↳ processing.
    ])
```

```
[52]: # Separate features and target.
X = data[features]
y = data[target]

# Split into training and testing sets (80% training, 20% testing).
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↳ random_state=42)
```

```
[53]: # Define model pipelines with preprocessing.
linear_model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                         ('regressor', LinearRegression())])

rf_model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                                     ('regressor',
                                      RandomForestRegressor(n_estimators=100, random_state=42))])

# Train the Linear Regression model.
linear_model_pipeline.fit(X_train, y_train)

# Train the Random Forest Regressor model.
rf_model_pipeline.fit(X_train, y_train)
```

```
[53]: Pipeline(steps=[('preprocessor',
                       ColumnTransformer(transformers=[('num', 'passthrough',
                                                         ['runtime', 'rt_rating']),
                                                         ('cat',
                                                          OneHotEncoder(drop='first',
                                                                           handle_unknown='ignore'),
                                                                           ['genre', 'director',
                                                                           'actors'])])),
                       ('regressor', RandomForestRegressor(random_state=42))])
```

```
[54]: # Function to evaluate model performance.
def evaluate_model(model, X_train, X_test, y_train, y_test):

    # Predictions on training and test sets.
    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    # Calculate metrics.
    metrics = {
        'MAE Train': mean_absolute_error(y_train, y_train_pred),
        'MAE Test': mean_absolute_error(y_test, y_test_pred),
        'MSE Train': mean_squared_error(y_train, y_train_pred),
        'MSE Test': mean_squared_error(y_test, y_test_pred),
        'R2 Train': r2_score(y_train, y_train_pred),
        'R2 Test': r2_score(y_test, y_test_pred)
    }
    return metrics

def print_metrics(title, metrics):
    print(f"{title}:")
    # Iterate through the metrics dictionary and print each metric with its
    # value.
    for metric, value in metrics.items():
```

```

        print(f" {metric}: {value:.4f}")
    print()

# Evaluate Linear Regression model.
lin_reg_metrics = evaluate_model(linear_model_pipeline, X_train, X_test,
    ↪y_train, y_test)

# Evaluate Random Forest Regressor model.
rf_reg_metrics = evaluate_model(rf_model_pipeline, X_train, X_test, y_train,
    ↪y_test)

# Print metrics for both models.
print_metrics("Linear Regression Performance", lin_reg_metrics)
print_metrics("Random Forest Regression Performance", rf_reg_metrics)

```

Linear Regression Performance:

```

MAE Train: 0.0000
MAE Test: 0.1819
MSE Train: 0.0000
MSE Test: 0.0598
R2 Train: 1.0000
R2 Test: 0.0534

```

Random Forest Regression Performance:

```

MAE Train: 0.0594
MAE Test: 0.1894
MSE Train: 0.0060
MSE Test: 0.0797
R2 Train: 0.8475
R2 Test: -0.2615

```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\preprocessing_encoders.py:242: UserWarning: Found unknown categories in columns [0, 1, 2] during transform. These unknown categories will be encoded as all zeros

```
warnings.warn(
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\preprocessing_encoders.py:242: UserWarning: Found unknown categories in columns [0, 1, 2] during transform. These unknown categories will be encoded as all zeros

```
warnings.warn(
```

During model evaluation, some categories appeared in the test data that were not present in the training data. The encoder handled these unknown categories by assigning them zero values. This warning does not stop the model from running, but it shows that the dataset contains categories that may not appear consistently across training and testing splits.

```
[55]: # Choose the model with the best test R-squared score.
def print_selected_model(selected_model, selected_metrics):
    print(f"Selected Model: {selected_model}")
    print("Performance Metrics:")
    # Iterate through the selected metrics and print each metric with its value.
    for metric, value in selected_metrics.items():
        print(f"  {metric}: {value:.4f}")
    print()
```

7 Milestone 3: Overview/Conclusion.

The best-performing model was **Linear Regression**, chosen based on its relatively better R-squared, MAE, and MSE performance on the test set than the Random Forest model.

Performance Metrics:

Mean Absolute Error (MAE): 0.1819

Mean Squared Error (MSE): 0.0598

R-squared (R^2): 0.0534

Insights: The results indicate that factors such as runtime, rt_rating, and other features predict imdbRating. However, the low R-squared score on the test set (0.0534) suggests that the Linear Regression model does not capture much of the variance in the test data despite performing nearly perfectly on the training set. This discrepancy may indicate overfitting, where the model has learned the training data too closely and fails to generalize well to new data.

While the Linear Regression model has a high R-squared score on the training set (almost 1), it may not effectively capture complex relationships within the data. This suggests that further tuning or possibly a different approach is needed for improved test performance.

Future Work: To improve accuracy, further steps could include additional feature engineering (e.g., extracting sentiment or keywords from the plot or Awards columns) or using more advanced models better at handling non-linear relationships.

Regularization techniques like Ridge or Lasso Regression could also help reduce overfitting and improve the model's generalization on the test set.

This concludes the model selection, building, and evaluation process for Milestone 3.

8 Additional Note:

For this milestone, I used a cleaned .csv file different from the file I submitted for Milestone 2.

9 Rewrite Milestone 3 and answer comments.

I had issues re-running my Milestone 3 file. I decided to rewrite the code.

```
[56]: # Import necessary libraries.
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from matplotlib import pyplot as plt

[57]: # Load the data.
data = pd.read_csv('cleaned_movie_data.csv')

# Convert 'imdbVotes' from object to integer.
data['imdbVotes'] = data['imdbVotes'].astype(str)
data['imdbVotes'] = data['imdbVotes'].str.replace(',', '').astype(int)

# Convert 'BoxOffice' from object to float.
data['BoxOffice'] = data['BoxOffice'].str.replace('$', '').str.replace(',', '').
    ↪astype(float)

# Convert 'rt_rating' from object to integer.
data['rt_rating'] = pd.to_numeric(data['rt_rating'].str.replace('%', ''),
    ↪errors='coerce').fillna(0).astype(int)

# Select relevant features and target.
features = ['genre', 'runtime', 'director', 'actors', 'rt_rating']
target = 'imdbRating'

# Drop rows with missing values in selected columns.
data = data[features + [target]].dropna()

[58]: # Define feature categories for preprocessing.
numeric_features = ['runtime', 'rt_rating']
categorical_features = ['genre', 'director', 'actors']

# Define the preprocessing pipeline.
numeric_transformer = 'passthrough' # Numeric features don't need scaling for
    ↪now
categorical_transformer = OneHotEncoder(drop='first', handle_unknown='ignore')

preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, numeric_features),
```

```
        ('cat', categorical_transformer, categorical_features)
    ])

```

```
[59]: # Separate features and target.
X = data[features]
y = data[target]

# Split into training and testing sets (80% training, 20% testing).
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
    ↪random_state=42)

```

```
[60]: # Define model pipelines with preprocessing.
linear_model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
    ('regressor', LinearRegression())])

rf_model_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
    ('regressor',
    ↪RandomForestRegressor(n_estimators=100, random_state=42))])

# Train the Linear Regression model.
linear_model_pipeline.fit(X_train, y_train)

# Train the Random Forest Regressor model.
rf_model_pipeline.fit(X_train, y_train)

```

```
[60]: Pipeline(steps=[('preprocessor',
    ColumnTransformer(transformers=[('num', 'passthrough',
    ['runtime', 'rt_rating']),
    ('cat',
    OneHotEncoder(drop='first',
    handle_unknown='ignore'),
    ['genre', 'director',
    'actors'])])),
    ('regressor', RandomForestRegressor(random_state=42))])

```

```
[61]: # Function to evaluate model performance with formatted output.
def evaluate_model(model, X_train, X_test, y_train, y_test):
    # Predictions on training and test sets.
    y_train_pred = model.predict(X_train)
    y_test_pred = model.predict(X_test)

    # Calculate metrics.
    metrics = {
        'MAE Train': mean_absolute_error(y_train, y_train_pred),
        'MAE Test': mean_absolute_error(y_test, y_test_pred),
        'MSE Train': mean_squared_error(y_train, y_train_pred),
        'MSE Test': mean_squared_error(y_test, y_test_pred),
    }

```

```

        'R2 Train': r2_score(y_train, y_train_pred),
        'R2 Test': r2_score(y_test, y_test_pred)
    }

    # Format the output to improve readability (2 decimal places for most
    ↪metrics).
    formatted_metrics = {key: f"{value:.4f}" for key, value in metrics.items()}

    # Return formatted metrics as a dictionary.
    return formatted_metrics

# Evaluate Linear Regression model.
lin_reg_metrics = evaluate_model(linear_model_pipeline, X_train, X_test,
    ↪y_train, y_test)

# Evaluate Random Forest Regressor model.
rf_reg_metrics = evaluate_model(rf_model_pipeline, X_train, X_test, y_train,
    ↪y_test)

# Combine the results into a single DataFrame, displaying metrics as columns.
metrics_df = pd.DataFrame({
    'Linear Regression': lin_reg_metrics,
    'Random Forest': rf_reg_metrics
})

# Display the results.
print("\nModel Comparison - Performance Metrics:")
print(metrics_df)

```

Model Comparison - Performance Metrics:

	Linear Regression	Random Forest
MAE Train	0.0000	0.0594
MAE Test	0.1819	0.1894
MSE Train	0.0000	0.0060
MSE Test	0.0598	0.0797
R2 Train	1.0000	0.8475
R2 Test	0.0534	-0.2615

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\preprocessing_encoders.py:242: UserWarning: Found unknown categories in columns [0, 1, 2] during transform. These unknown categories will be encoded as all zeros

warnings.warn(
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\preprocessing_encoders.py:242: UserWarning: Found unknown categories in columns [0, 1, 2] during transform. These unknown categories will be encoded as all zeros

```
warnings.warn(
```

Note: The model produced a warning because some categories appeared in the test data that were not present in the training data. The encoder handled these unknown categories by encoding them as zeros. This does not stop the model from running, but it shows that some movie categories, such as genres, directors, or cast values, may not appear consistently across the training and testing sets.

```
[62]: # Choose the model with the best test R-squared score.
if float(rf_reg_metrics['R2 Test']) > float(lin_reg_metrics['R2 Test']):
    selected_model = 'Random Forest'
    selected_metrics = rf_reg_metrics
else:
    selected_model = 'Linear Regression'
    selected_metrics = lin_reg_metrics

# Display the selected model and performance metrics in a more readable format.
print(f"Selected Model: {selected_model}")

# Convert the selected model's metrics to a pandas DataFrame and display as a
↳column.
selected_metrics_df = pd.DataFrame(selected_metrics.items(), columns=['Metric',
↳'Value'])

# Format the "Value" column to 4 decimal places for better readability.
def format_value(value):
    try:
        # Try to format the value as a float with 4 decimal places.
        return f"{float(value):.4f}"
    except ValueError:
        # If it's not a number (e.g., string), return it as is
        return value

# Apply the formatting function to the "Value" column.
selected_metrics_df['Value'] = selected_metrics_df['Value'].apply(format_value)

# Display the performance metrics of the selected model.
print("\nPerformance Metrics:")
print(selected_metrics_df)
```

Selected Model: Linear Regression

Performance Metrics:

	Metric	Value
0	MAE Train	0.0000
1	MAE Test	0.1819
2	MSE Train	0.0000
3	MSE Test	0.0598

```
4 R2 Train 1.0000
5 R2 Test 0.0534
```

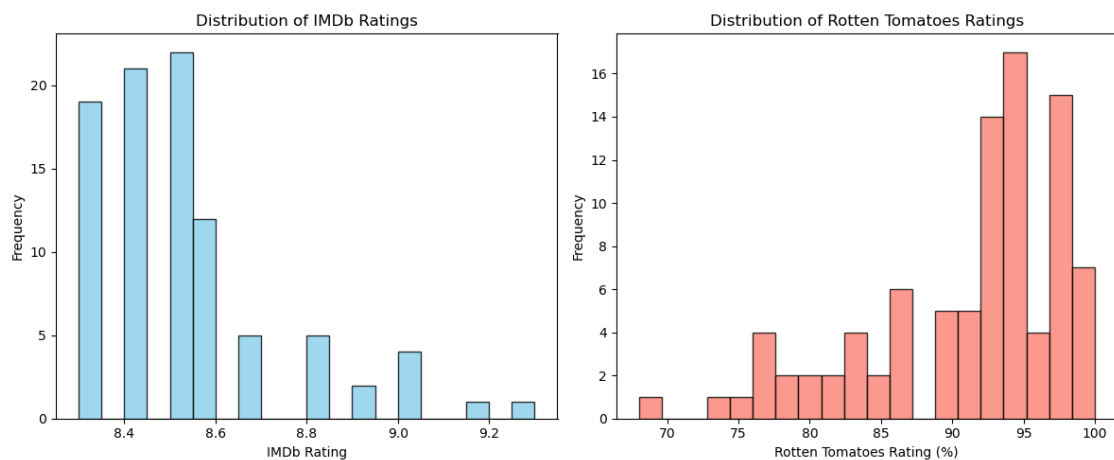
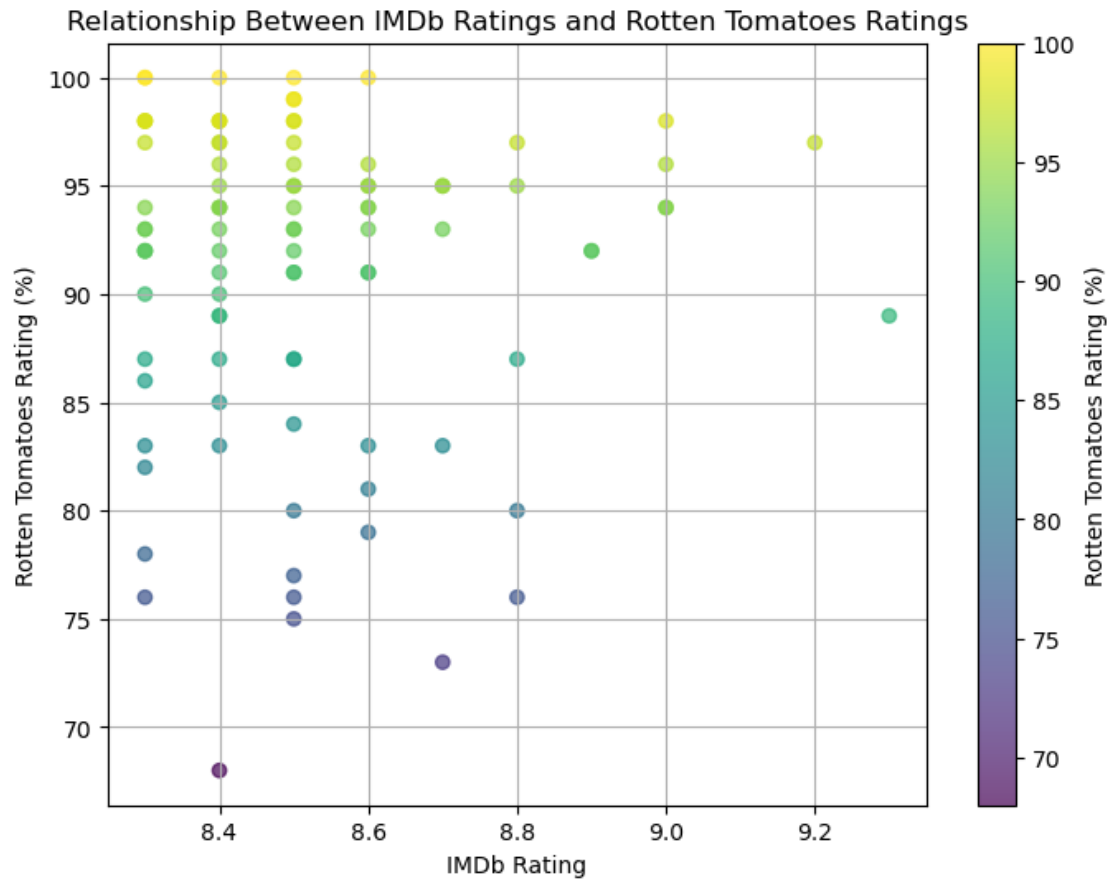
```
[63]: # Scatter plot for better visualization.
plt.figure(figsize=(8, 6))
plt.scatter(data['imdbRating'], data['rt_rating'],
            c=data['rt_rating'], cmap='viridis', alpha=0.7)
plt.colorbar(label='Rotten Tomatoes Rating (%)')
plt.title('Relationship Between IMDb Ratings and Rotten Tomatoes Ratings')
plt.xlabel('IMDb Rating')
plt.ylabel('Rotten Tomatoes Rating (%)')
plt.grid(True)
plt.show()

# Histograms.
plt.figure(figsize=(12, 5))

# IMDb Rating Distribution.
plt.subplot(1, 2, 1)
plt.hist(data['imdbRating'], bins=20, edgecolor='black', alpha=0.8,
        color='skyblue')
plt.title('Distribution of IMDb Ratings')
plt.xlabel('IMDb Rating')
plt.ylabel('Frequency')

# Rotten Tomatoes Rating Distribution.
plt.subplot(1, 2, 2)
plt.hist(data['rt_rating'], bins=20, edgecolor='black', alpha=0.8,
        color='salmon')
plt.title('Distribution of Rotten Tomatoes Ratings')
plt.xlabel('Rotten Tomatoes Rating (%)')
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```



Scatter Plot Findings The scatter plot shows that there is no strong connection between IMDb ratings and Rotten Tomatoes ratings. The very low correlation ($r = 0.011$) confirms this, meaning that how a movie is rated on one platform doesn't seem to influence its rating on the other. This

suggests that the two platforms likely use different ways to evaluate movies or appeal to various groups of people.

Histogram Insights Looking at the distributions of ratings: - **IMDb Ratings:** Most ratings are tightly packed between 8.0 and 9.5, showing that IMDb users tend to give higher ratings overall. This could mean that people on IMDb mostly rate movies they already like or expect to enjoy. - **Rotten Tomatoes Ratings:** Ratings are spread out more evenly, with noticeable peaks around 80% and 90%. This may be because Rotten Tomatoes combines scores from moviegoers and professional critics, introducing more variety.

What This Means The data suggests that IMDb and Rotten Tomatoes approach movie ratings in very different ways: - IMDb focuses on user reviews, which might reflect personal preferences or fandom. - Rotten Tomatoes combines audience scores and critics' reviews, leading to a broader range of ratings.

9.0.1 Evaluation

The lack of a clear relationship between the two rating systems highlights their unique audiences and rating styles. IMDb might be better for understanding fan enthusiasm, while Rotten Tomatoes offers a mix of public opinion and expert feedback. This difference helps explain why their ratings don't match up.

9.0.2 Next Steps

To better understand the differences between the platforms, future analysis could: 1. **Use the Same Scale:** Adjust both ratings to a 0–100 scale for easier comparison. 2. **Check for Outliers:** Look at extreme ratings to see if they affect the results. 3. **Find Other Patterns:** See if specific genres or other factors explain the differences.

These steps could provide more insight into how these two platforms measure a movie's success.